2026 prelim electricity answers

Answer pages for the 2026 prelim electricity and magnetism questions.
Whole pages are kept. A page that also holds another answer is included in full.
2026 RI has no answer scheme in the corpus, so RI is not in this file.
DHS, NYJC and VJC answers are the worked papers. EJC and HCI answers are solution booklets.

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4 A technician is designing a charging circuit for a compact camera flash unit as shown in Fig. 4.1. The circuit consists of a DC battery V of 12 V, a resistor R = 15 kΩ and an uncharged capacitor C of 220 μF, connected in series with a switch.

At time t = 0, the switch is closed.

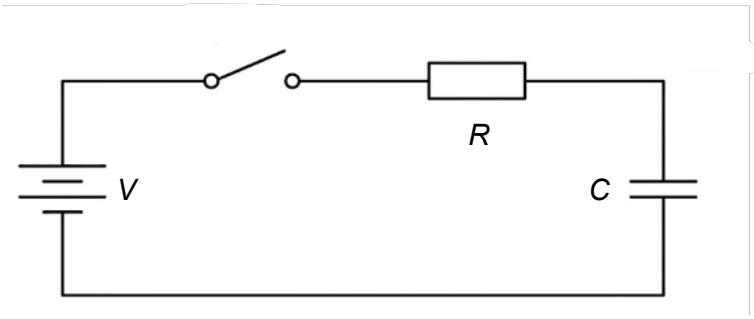


Fig. 4.1

(a) Calculate the time constant for the circuit.

The time constant of a series RC circuit is the product of the resistance and the capacitance:

$RC = 15 \times 10^3 \times 220 \times 10^{-6} = 3.3$ s [A1]

time constant = ... s [1]

(b) Calculate the percentage of maximum charge in the capacitor 6.6 s after the switch is closed.

$$Q = Q_0\left(1 - e^{-\frac{t}{RC}}\right) \Rightarrow \frac{Q}{Q_0} = \left(1 - e^{-\frac{6.6}{3.3}}\right) \quad \text{[B1]}$$

$$= 0.8647 \approx 87\% \quad \text{[A1]}$$

percentage = ... % [2]

(c) Hence or otherwise, determine the current flowing through the circuit exactly 6.6 s after the switch is closed.

$$Q \propto V_C \Rightarrow \frac{V_C}{V_0} = 0.8647 \quad \text{[B1]}$$

$$V_C = 0.8647(12) = 10.376 \text{ V}$$

$$V_R = 12 - V_C = 1.624 \text{ V}$$

$$I = \frac{V_R}{R} = \frac{1.624 \text{ V}}{15 \times 10^3\ \Omega} = 0.108 \text{ mA} \quad \text{[A1]}$$

alternatively:

charging current, $I = I_0 e^{-\frac{t}{RC}}$

$$I = \frac{V_0}{R}\left(e^{-\frac{6.6}{3.3}}\right) = 0.108 \text{ mA}$$

current = ... mA [2]

[Total: 5]

5 A circuit is set up as shown in Fig. 5.1. The circuit contains a battery of e.m.f. 12 V, and a fixed resistor r of resistance 6 Ω and length L_r = 0.50 m. A movable metal rod M, of length L_M = 0.60 m and resistance 12 Ω, is placed on a conducting track at a distance d = 4.0 cm from r. The conducting track is assumed to be smooth and has zero resistance.

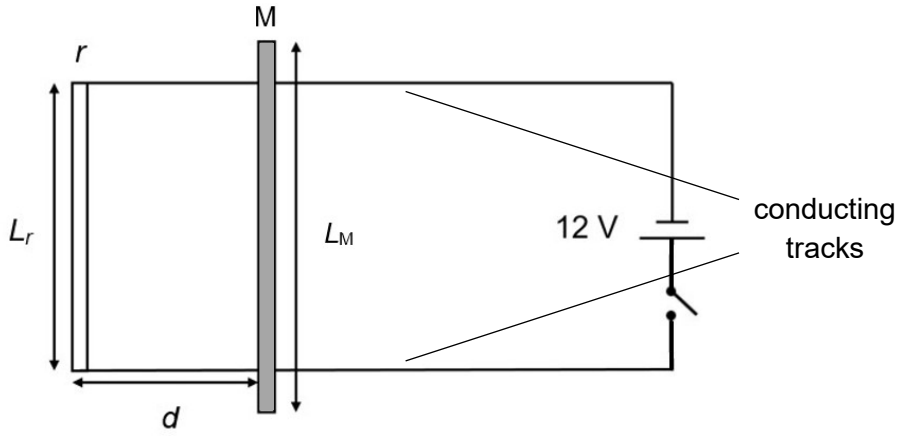


Fig. 5.1 (not to scale)

When the switch is now closed.

(a) Show that the magnetic flux density B at the metal rod M due to the current in resistor r is 10 μT.

the current passing through r, $I_r = \frac{12\text{ V}}{6\ \Omega} = 2.0\text{ A}$

$B = \frac{\mu_0 I_r}{2\pi d} = \frac{(4\pi \times 10^{-7}) \times 2.0}{2\pi(0.040)} = 1.0 \times 10^{-5}\text{ T} = 10\ \mu\text{T}$ (shown) [B1]

[1]

(b) Calculate the magnetic force experienced by M and state its direction.

$I_M = \frac{12\text{ V}}{(\frac{0.50}{0.60})(12\ \Omega)} = 1.2\text{ A}$

The magnetic force on M, acting over the current-carrying length of 0.50 m within the field of r:

$F = BI_M L = (1.0 \times 10^{-5})(1.2)(0.50)$ [M1]

$= 6.0 \times 10^{-6}$ N [A1]

Direction of F: leftwards, from Fleming's left-hand rule. [B1]

magnetic force = ... N

direction: ..... [3]

(c) M in **(b)** is held stationary and a magnetic field of flux density 1.0 mT is now directed out of page throughout the circuit in Fig. 5.1. M is now released.
Describe and explain the initial motion of M.

M moves and accelerates to the right. [B1]

The **applied magnetic field of 1.0 mT is much larger than 10 μT produced by r**. By Fleming's left-hand rule, the **applied field exerts a force to the right on M, which is larger than the 6.0 × 10^{-6} N force to the left due to the field of** *r*. Hence, the net force on M is to the right. [B1]

..... [2]

(d) A proton enters the magnetic field *B* at an angle to the field of less than 90°.
The proton follows the path shown in Fig. 5.2.

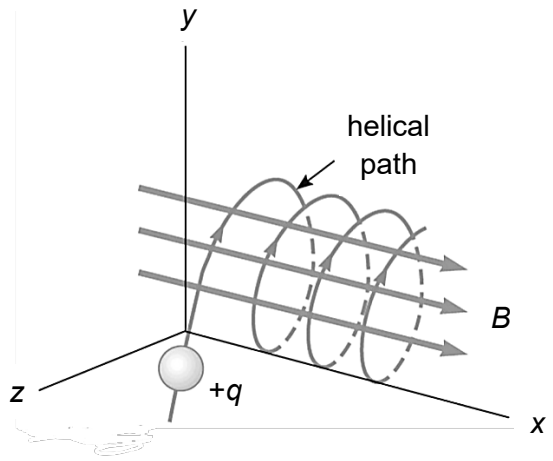


Fig. 5.2

(i) Explain why the proton follows this helical path.

Since the proton enters at an angle to *B*, its velocity has two components: **one perpendicular to *B* and one parallel to *B***. [B1]

The perpendicular component **experiences a magnetic force** ($F = Bqv\sin\theta$) that is always **perpendicular to its velocity**, **providing the centripetal force** for uniform circular motion. The parallel component experiences **no magnetic force** and so the proton continues to move along the direction of *B* at **constant speed**. The combination of these two motions produces a helical path. [B1]

.....

.....

..... [2]

(ii) Fig. 5.3 shows a magnetic bottle, where a non-uniform magnetic field is produced between two current carrying coils. A proton is initially directed perpendicularly into the page with speed *v* at the position shown.

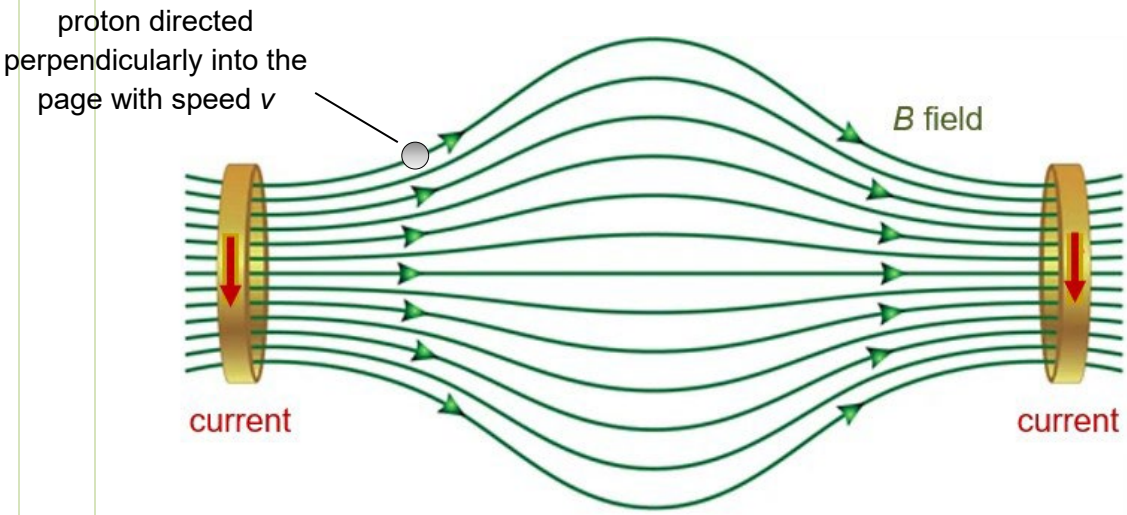


Fig. 5.3

Describe and explain the subsequent motion of the proton in the magnetic bottle, assuming the proton remains within the field region throughout.

The proton will experience a magnetic force perpendicular to the magnetic field based on Flemming's left hand rule. This **magnetic force, pointing towards the centre of the magnetic bottle**, produces velocity parallel and perpendicular to the magnetic field. **Past the centre** of the magnetic bottle, the magnetic force perpendicular to the field, **pointing towards the centre of the magnetic bottle**, opposes the velocity parallel to magnetic field, (reducing its magnitude and changes the direction of velocity eventually) [B1*]

This combined motion of a velocity parallel and perpendicular to the magnetic field in the non-uniform magnetic field produces a **spiral motion** from **one end of the magnetic bottle to the other** where the **radius of the motion increases towards the middle and decreases towards the ends**.[B1]

This process continues and the proton oscillate indefinitely between the two coils, while spiralling around the axis of the bottle.

[To take note] Throughout the motion, since the magnetic force is always perpendicular to the velocity, it does no work on the proton and the speed of the proton remains constant — only the distribution between the velocity components parallel and perpendicular to the B field such that the kinetic energy remains constant.

[Total: 10]

7 Read the passage and answer the questions that follow.

Traditional wind turbines are limited by their height; the friction of the Earth's surface slows wind speeds at low altitudes, and the massive concrete foundations required for towers over 150 metres tall are both expensive and environmentally invasive. To overcome these limitations, engineers have developed **Airborne Wind Energy Systems (AWES)**.

In early 2026, the S2000 Stratosphere Airborne Wind Energy System (SAWES), developed in China, became the world's first megawatt-class "flying power plant" to successfully feed electricity into a national grid. The system utilizes a helium-filled, ring-shaped aerostat (a "balloon") anchored to a ground station by a high-strength tether. A simplified schematic of the S2000 system is shown in Fig. 7.1. The aerostat remains aloft through upthrust, the upward force exerted by the surrounding air on the balloon when it is immersed in the air. This allows the turbines to access the "jet effect" of the stratosphere—a region where winds are faster, more stable, and carry significantly more kinetic energy than those at ground level.

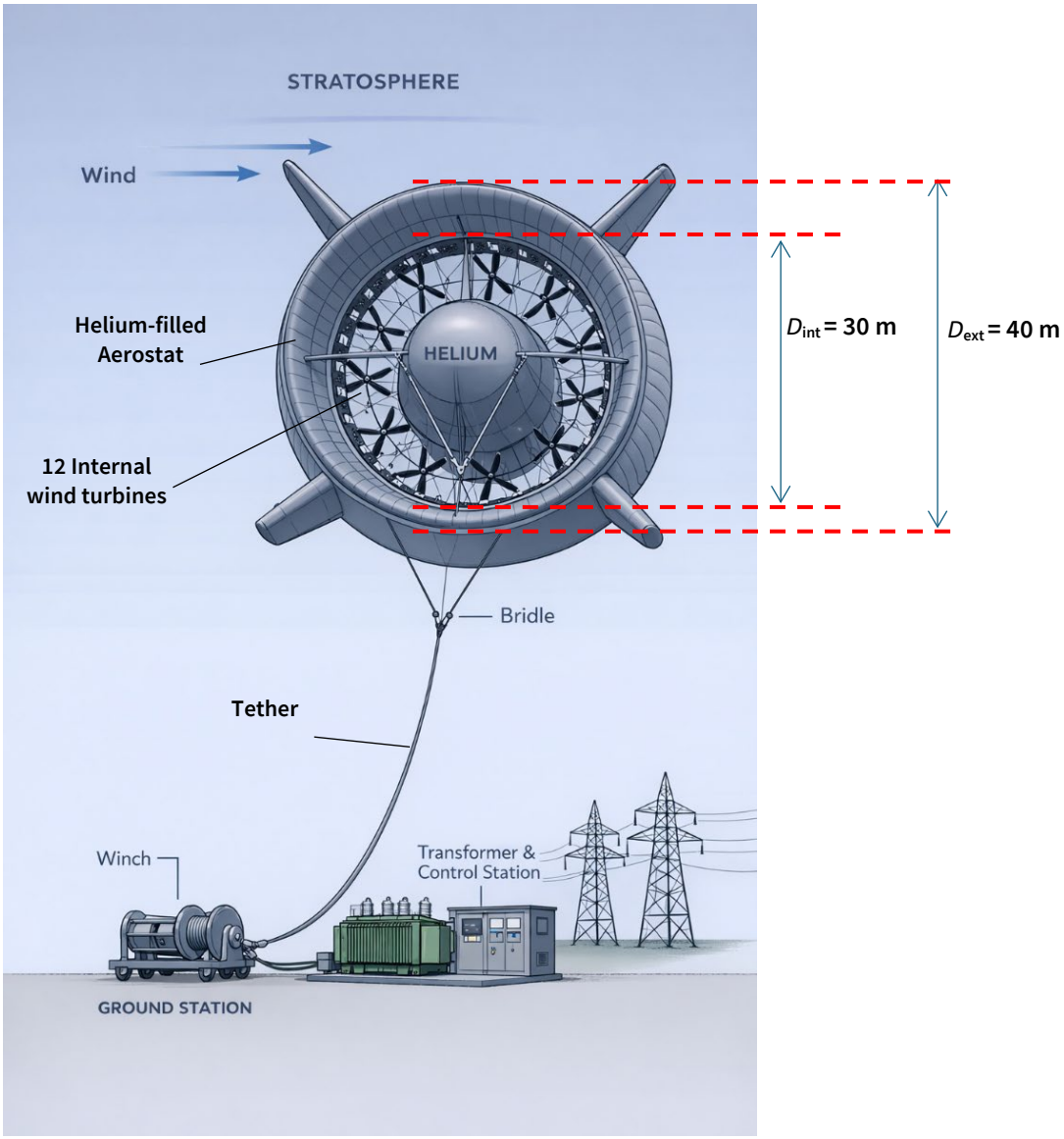


Fig. 7.1: Simplified Schematic of the S2000 System

The S2000 is linked to its ground-based winch station by a 2.5 km multipurpose conductive tether. This cable is a composite structure, featuring a high-tensile carbon-fibre core to withstand buoyancy and wind-drag forces, surrounding an aluminium-based conductive layer for power transmission. Onboard sensors relay real-time data through fibre-optic filaments within the tether, allowing a ground-based AI to adjust turbine speeds and aerostat pitch.

Table 7.1: S2000 SAWES System Specifications and Operating Parameters

Category	Parameter	Value (at h = 2000 m)
Atmospheric	Air density (ρ_{air})	1.007 kg m^{-3}
	Average Wind Speed (v)	15.0 m s^{-1}
Aerostat	External Diameter (D_{ext})	40 m
	Internal Diameter (D_{int})	30 m
	Volume of Helium (V_{He})	20,000 m^3
	Aerostat Structural Mass ($m_{structure}$)	5,000 kg
Generation	Total Turbine Swept Area ($A_{turbine}$)*	340 m^2
	Turbine Assembly Mass ($m_{turbine}$)*	2,400 kg
	Maximum Rated Power (P_{max})	3.0 MW
	Average Power Output from Initial Test	770 kW (over 30 min)
Tether	Total Length l_{tether}	2500 m
	Linear Mass	1.8 kg m^{-1}

*Note: Total of the 12 internal turbines.

Additional Constants and Parameters:

Aluminium resistivity: $\rho_{Al} = 2.8 \times 10^{-8}\ \Omega$ m

Helium density: $\rho_{He} =$ 0.18 kg m^{-3}

Power coefficient: $C_p = 0.40$ (maximum)

System efficiency: $\eta_{system} = 45\%$ (maximum)

Drag coefficient (ring aerostat): $C_d = 0.25$

Energy density of the stratospheric wind, which represents the kinetic energy per unit volume of moving air, helps engineers assess the energy resource available at different altitudes and locations. This, in turn, will help to determine the amount of power that can be harnessed from the wind through the turbine.

Real turbines cannot extract all available wind power. The power coefficient (C_p) represents the fraction of wind power actually converted to mechanical power. For example, a C_p of 0.40 means the turbine converts 40% of the wind's kinetic energy into mechanical rotation. System efficiency (η_{system}) accounts for subsequent losses during electrical conversion and transmission, reducing the final delivered power. This includes losses in the generator, power electronics, and the resistance of the 2.5 km tether. A system efficiency of 45% means only 45% of the mechanical power extracted by the turbines reaches the ground station as usable electricity.

(a)	**(i)**	Using data from Table 7.1, calculate the buoyant force (upthrust) $\rho_{air} V_{He} g$ acting on the helium-filled aerostat at its operating altitude.
		$\rho_{air} V_{He} g = 1.007 \times 20{,}000 \times 9.81 = 197.57 \times 10^3$ N [A1]
		buoyant force = ... N [1]
	(ii)	Calculate the total weight of Helium-filled aerostat.
		Weight of helium = 0.18 × 20,000 × 9.81 = 35,316 N Weight of aerostat = 5,000 × 9.81 = 49,050 N Weight of turbines = 2,400 × 9.81 = 23,544 N Total weight = 107,910 N [A1] = 108, 000 N
		total weight = ... N [2]
	(iii)	Using your answers from **(a)(i)** and **(a)(ii)**, determine the minimum tension in the tether.
		minimum tension = 197,570 − 107,910 = 89,660 N = 90,000 N [A1]
		minimum tension = ... N [1]

	(iv)	Calculate the weight of the tether.
		weight of tether = 2,500 × 1.8 × 9.81 = 44,145 = 44,000 N
		weight = ……. N [1]
	(v)	State whether the force exerted by the Winch on the tether will be greater, equal or smaller than your answer in **(a)(iii)**, explain your answer.
		The force exerted by the Winch on the tether will be smaller. [A1] Since the tether is in equilibrium, (considering the forces on the tether), the force on the tether upwards by the aerostat [answer in (a)(iii)] is equal to the weight of the tether and the force exerted by the Winch on the tether (downwards) [M1] ……..…. [2]
(b)	**(i)**	When wind flows around the ring-shaped aerostat, it experiences a drag force given by $\frac{1}{2}\rho v^2 C_d A$, where A is the frontal area presented to the wind. Assuming the frontal area is a circle with external diameter D_{ext}, calculate the drag force acting on the aerostat in the stratospheric wind conditions.
		Frontal area = $\frac{\pi D_{ext}^2}{4} = \frac{\pi(40^2)}{4} = 1256.6\ \text{m}^2$ [A1] Drag force, $F_D = \frac{1}{2}\rho v^2 C_d A = \frac{1}{2}(1.007)(15.0)^2(0.25)(1256.6) = 35590\ \text{N} \approx 36\ \text{kN}$ [A1] [Both the frontal area calculation and the substitution into the drag force formula must be shown explicitly. Award zero if working is absent even if the answer is correct.]
		drag force = ……. N [2]
	(ii)	Determine the magnitude of the change in tension in the tether, when drag force acts on the aerostat. Explain your answer.
		Magnitude of the change in tension in the tether = 36 kN [A1] Tension is a vector, a change in tension $\Delta \boldsymbol{T}$ will be vector subtraction given by $(\boldsymbol{T}_f - \boldsymbol{T}_i)$, which will be the opposite of the drag force. Therefore, magnitude of the change in tension $\lvert\Delta \boldsymbol{T}\rvert$, is 36 kN. [B1] Alternatively, since $\boldsymbol{T}_f = \boldsymbol{T}_i - \boldsymbol{D}$, we have $\boldsymbol{T}_f - \boldsymbol{T}_i = -\boldsymbol{D}$, which implies that the drag force D causes the change in tension in the tether.
		magnitude of change in tension = ……. N [2]

(c)	**(i)**	By considering the energy density of the stratospheric wind, show that the power that can be harnessed from the wind can be expressed as $\frac{1}{2}\rho_{air}A_{turbine}v^3$, where ρ_{air}, $A_{turbine}$ and v are quantities expressed in Table 7.1.
		energy density $= \left(\frac{1}{2}mv^2\right)_{air} \Big/ V_{air} = \frac{1}{2}\rho_{air}v^2$ [A1] $P = (\text{energy density})(\text{air volume change rate}) = \left(\frac{1}{2}\rho_{air}v^2\right)(A_{turbine}v) = \frac{1}{2}\rho_{air}A_{turbine}v^3$
		[2]
	(ii)	Determine the total power harnessed from the wind through the turbine.
		$P = \frac{1}{2}\rho_{air}A_{turbine}v^3 = \frac{1}{2}(1.007)(340)(15.0)^3$ $= 577.8\times10^3$ W [A1]
		total power = ... W [1]
	(iii)	Hence or otherwise, determine maximum electrical power delivered to ground.
		Mechanical power = $C_p \times P = 0.40\times577.8\times10^3 = 231.1\times10^3$ W [B1] Delivered power = $\eta_{system}\times231.1\times10^3 = 0.45\times231.1\times10^3 = 104.0\times10^3$ W [A1]
		power = ... W [2]
	(iv)	The average power output from the Initial test from Table 7.1 is 770 kW. State whether this is higher or lower than your calculated value in **(c)(iii)** and suggest a reason for the difference.
		Initial test shows 770 kW average (from Table 7.1), much higher than calculated value. [B1] Difference suggests variable wind conditions with periods of higher wind speeds during 30 min testing period. [B1] $\left\{P = (\text{energy density})(A_{turbine}v) = \frac{1}{2}\rho_{air}A_{turbine}v^3\right\}$ ..... [2]

(d)	Dimensions of the tether must be considered in order to carry a current of up to 300 A and heat losses through the tether only account for 0.48% of the maximum rated power $\left(P_{max}\right)$ for efficient transmission of power. Determine the cross-sectional area of the aluminium needed in the conductive layer of the tether that satisfies the above considerations.	
	$\frac{0.48}{100} = \frac{P_{loss}}{P_{max}} \Rightarrow P_{loss} = \frac{0.48}{100} \times \left(3.0 \times 10^6\right) = 14.4 \text{ kW}$ $P_{loss} = I^2 R \Rightarrow R = \frac{P_{loss}}{I^2} = \frac{14400}{300^2} = 0.16\ \Omega$ $R = \rho_{Al} \frac{l_{tether}}{A_{tether}}$ $\Rightarrow 0.16 = 2.8 \times 10^{-8} \left(\frac{2500}{A_{tether}}\right)$ $\therefore A_{tether} = 4.4 \times 10^{-4} \text{ m}^2$	
	cross-sectional area = …… m^2 [2]	
		[Total: 20]

Section A

Answer **all** the questions in this Section in the spaces provided.

1 A battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance is connected in series with two fixed resistors and a thermistor, as shown in Fig. 1.1.

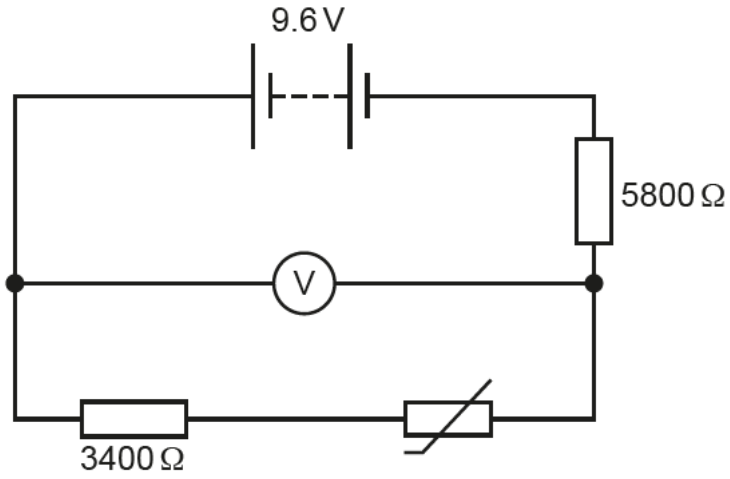


Fig. 1.1

The fixed resistors have resistances of 3400 Ω and 5800 Ω. The reading on the voltmeter in the circuit is 6.0 V.

(a) Calculate the current in the resistor of resistance 5800 Ω.

$9.6 = 6.0 + (I \times 5800)$ or $3.6 = I \times 5800$ [C1]

$I = 6.2 \times 10^{-4}$ A [A1]

current = ... A [2]

(b) Calculate the resistance of the thermistor.

$9.6 = 6.2 \times 10^{-4} \times (3400 + 5800 + R)$ [C1]

or

$6.0 = 6.2 \times 10^{-4} \times (3400 + R)$

$R = 6.3 \times 10^{3}\ \Omega$ [A1]

resistance = ... Ω [2]

(c) The initial energy stored in the battery is 2.6×10^4 J.

Assume that the e.m.f. of the battery is constant.

Determine the final energy stored in the battery after a charge of 330 C has moved through it.

$(\Delta W = \varepsilon \times Q =) 9.6 \times 330 (= 3170 \text{ J})$ [C1]

final stored energy = $2.6 \times 10^4 - 3170$
$= 2.3 \times 10^4$ J [A1]

final stored energy = ... J [2]

(d) The environmental conditions change causing an increase in the resistance of the thermistor.

State whether there is a decrease, increase or no change to:

(i) the temperature of the thermistor

decrease [B1]
(since an NTC thermistor increases in resistance when temperature falls) . [1]

(ii) the current in the thermistor

decrease [B1]
. (since total resistance increases with the e.m.f. fixed) ... [1]

(iii) the potential difference across the thermistor.

increase [B1]
... (p.d. across the fixed resistors decreases, and since the total e.m.f. is fixed, the p.d. across the thermistor increases) ... [1]

(e) If the temperature of the thermistor in Fig. 1.1 changes at a constant rate, state a reason why the potential difference across the thermistor does not change at a constant rate.

.. change of resistance with temperature is not linear (Thermistor is a non-ohmic conductor) [B1] .....

OR

.. potential divider equation is non-linear .....

..... [1]

[Total: 10]

5 Two point charges, A and B, are separated in a vacuum by a distance of 0.0100 cm, as shown in Fig. 5.1.

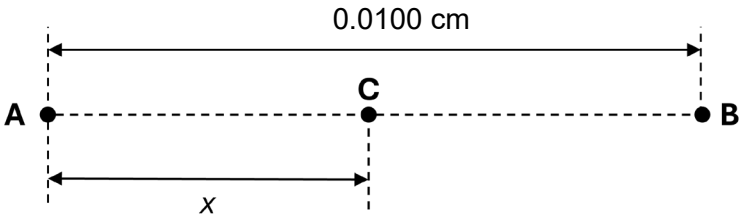


Fig. 5.1

A point C is on the line joining the two charges and is a distance x from charge A.

The variation with distance x of the electric potential V_x at point C is shown in Fig. 5.2.

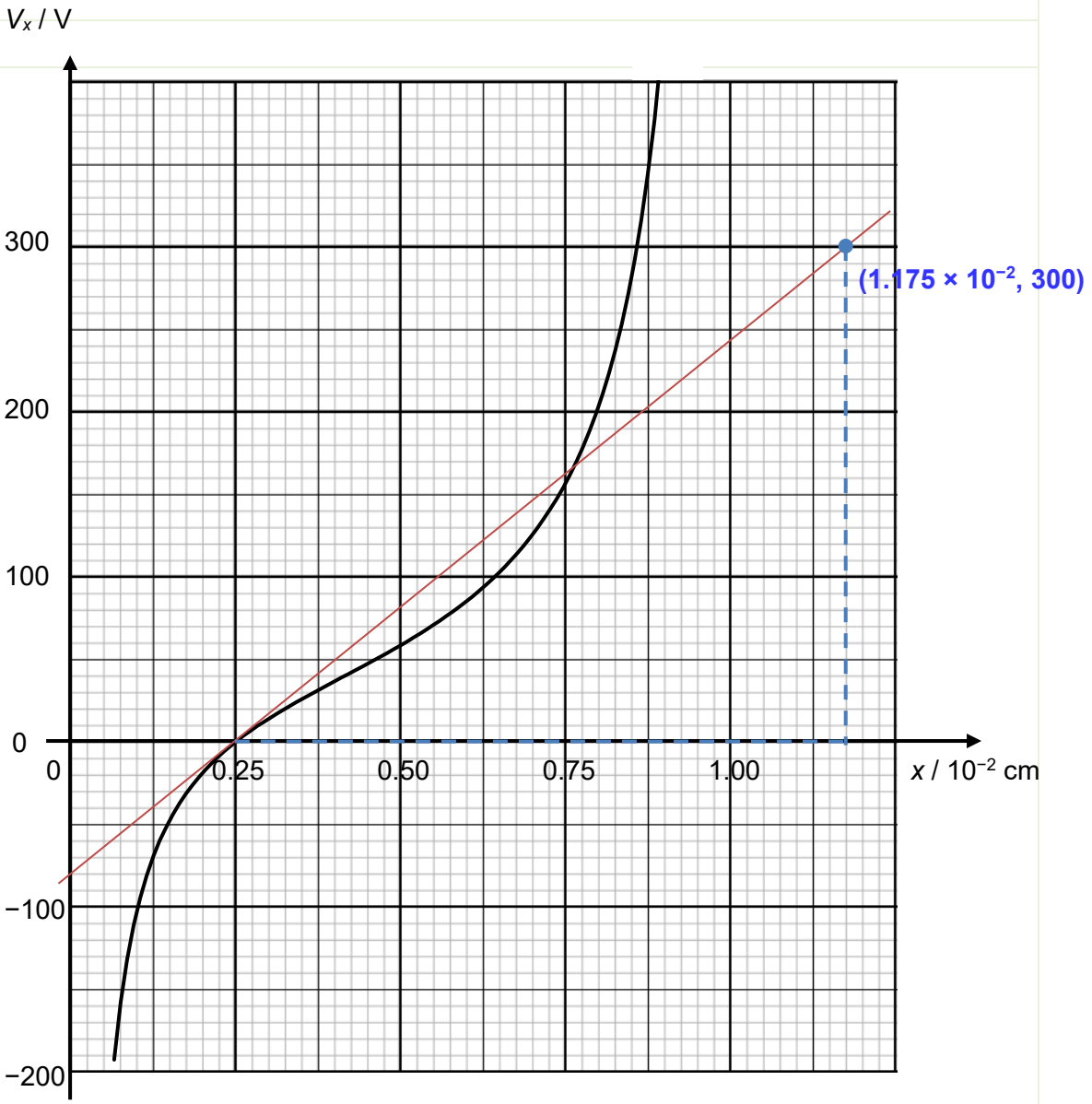


Fig. 5.2

(a) Define electric field strength at a point.

Electric field strength at a point in an electric field is defined as the electric force per unit positive charge exerted on a small test charge placed at that point. [B1]

..... [1]

(b) Explain whether the charges have the same or opposite signs.

The graph shows that the electric potential is zero at a point between the two charges (at x = 0.0025 cm), which is only possible if the two charges are of opposite signs — a positive charge produces a positive potential whilst a negative charge produces a negative potential. [2]

OR

Furthermore, since electric field strength is equal to the negative potential gradient, and the potential decreases continuously from B to A, the resultant electric field points from B towards A at every point between the charges. For this to be the case, the fields due to each individual charge must reinforce rather than cancel, which is only possible if the charges are of opposite signs. [2]

..... [2]

(c) Calculate the electric field strength where $x = 0.25 \times 10^{-2}$ cm.

$$\frac{dV}{dx} = \frac{(300-0)\text{ V}}{(1.175-0)\times10^{-4}\text{ m}} = 2.6\times10^{6}\text{ V m}^{-1} \quad \text{[M1]}$$

$$E = -\frac{dV}{dx}$$

$$= -2.6\times10^{6}\text{ V m}^{-1} \quad \text{[A1]}$$

Negative sign indicates E points in the direction of decreasing V. Therefore, the E has magnitude 2.6×10^{6} V m^{-1} and points in the negative x-direction (leftwards, towards charge A). [B1]

electric field strength = ... V m^{-1}

direction of electric field strength: ..... [3]

(d) An electron is projected with a speed of 7.73×10^6 m s^{-1} from the position where $x = 0.725 \times 10^{-2}$ cm towards point charge A.

Calculate the speed of the electron at the position where $x = 0.175 \times 10^{-2}$ cm.

$\Delta E_k = -\Delta E_p$ {conservation of mechanical energy}

$\Delta E_p = q(V_f - V_i)$ [M1]

$= e[(-30) - (140)]$

$= (-1.60 \times 10^{-19})(-170) = 2.72 \times 10^{-17}$ J

$\therefore \Delta E_k = \frac{1}{2} m_e [v_f^2 - v_i^2] = -2.72 \times 10^{-17}$ J [M1]

$\Rightarrow v_f = \sqrt{\frac{2\Delta E_k}{m_e} + v_i^2} = \sqrt{\frac{2(-2.72 \times 10^{-17})}{9.11 \times 10^{-31}} + (7.73 \times 10^6)^2}$ **=0.0 ms^{-1}** [A1]

speed = ... m s^{-1} [3]

[Total: 9]

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6 A simple ideal iron-cored transformer is illustrated in Fig. 6.1.
An alternating input p.d. V_p of 240 V is connected across the primary coil, of 120 turns of the transformer. The secondary p.d. V_s of 3600 V is applied to a load resistor.

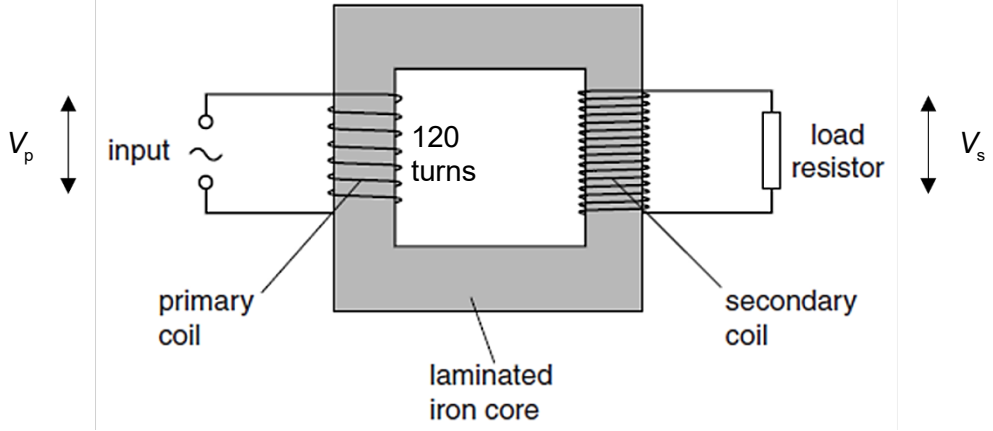


Fig. 6.1

(a) (i) Explain why an iron core is preferred over an air core in the construction of a transformer.

. it increases the magnetic flux density OR concentrates the magnetic flux as it links it from the primary to the secondary coil, (thereby ……
. increase the efficiency of energy transfer) [B1] ……

…… [1]

(ii) Explain why the iron core is laminated.

. to reduce energy losses by reducing eddy (induced) currents [B1] …….

……

…… [1]

(b) Calculate the number of turns in the secondary coil.

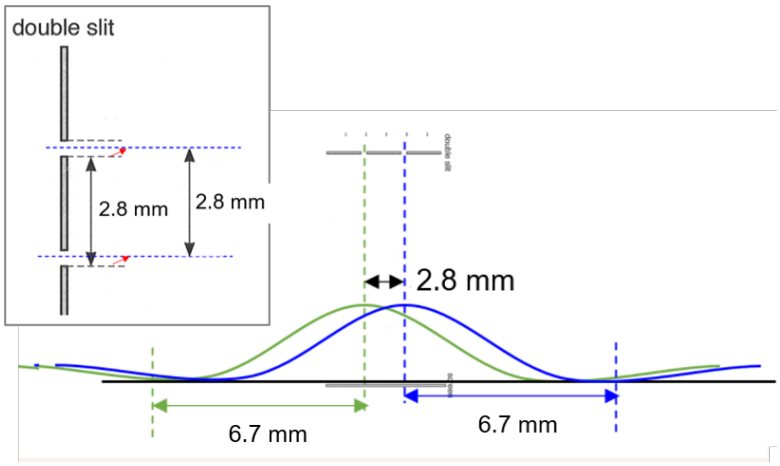
$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow N_s = 120\left(\frac{3600}{240}\right) = 1800 \quad \text{[B1]}$$

number of turns = ... [1]

	(c)	Using Faraday's Law, explain why an alternating current in the primary coil gives rise to an alternating e.m.f. in the secondary coil.
		The alternating current in the primary coil produces a **magnetic flux that changes continuously** in both magnitude and direction. [B1] Since the secondary coil is wound on the same iron core, **this changing magnetic flux links the secondary coil**, so the flux through the secondary coil also alternates. [B1] By Faraday's Law, the induced e.m.f. is proportional to the rate of change of magnetic flux linkage [$\varepsilon = -N(d\Phi/dt)$]. Since the flux is continuously changing, an e.m.f. is continuously induced in the secondary coil. **As the flux alternates, the rate of change also alternates in sign,** so the induced e.m.f. alternates as well. [B1] …… …… …… …… …… [3]
	(d)	A step-up transformer is used when transmitting electricity over long distances. Suggest a reason for this.
		…… A step-up transformer increases the voltage and reduces the current, thereby reducing the power loss in the transmission cables ($P = I^2R$), making transmission more efficient. [B1] …… …… …… …… [1]
		[Total: 7]

Qns	Answer	Marks
	$K = \frac{\frac{1}{2}m\Delta(v^2)}{t} = \frac{1}{2}\left(\frac{m}{t}\right)\left(v_f^{\,2} - v_i^{\,2}\right)$ $= \frac{1}{2}\left(\frac{100}{1000}\right)\left(\left(\frac{5.0}{100}\right)^2 - 0\right)$ $= 1.25 \times 10^{-4}\,\text{W}$	**B1**
2(b)(iv)	Perfectly inelastic collision. Energy transferred mechanically via work done by the motor is to the (horizontal)l kinetic store of the sand and to the internal store of the system (heating/friction).	**B1** **B1**
	TOTAL FOR QUESTION 2	**8**
3(a)	Electric potential at a point is the work done per unit charge by an external force in bringing a small positive test charge from infinity to that point.	**B1**
3(b)(i)	$E = -\frac{dV}{dr}$ $\lvert E \rvert$ = gradient of graph $= \left\lvert \frac{2.50 - 0}{(0 - 2.50) \times 10^{-2}} \right\rvert$ [M1] $= 100\,\text{N C}^{-1}$ [A1] Acceptable range: 88 <=E<= 1105□□ Direction: in positive x-direction / towards right / from left to right / → [B1]	**M1** **A1** **B1**
3(b)(ii)	The graph is symmetrical.	**B1**
3(b)(iii)	Using a point such as x = 0.20 cm where V ≈ 6.0 V: $V = \frac{Q}{4\pi\varepsilon_o x} + \frac{Q}{4\pi\varepsilon_o (0.040 - x)}$ $6.0 = \frac{Q}{4\pi\varepsilon_o}\left(\frac{1}{0.002} + \frac{1}{0.040 - 0.002}\right)$ [M1] $Q = 1.27 \times 10^{-12}\,\text{C}$ [A1] Accept also if student used other suitable points for substitution.	**M1** **A1**
	TOTAL FOR QUESTION 3	**7**
4(a)	Resistance of a circuit component is the ratio of the potential difference across the component to the current passing through it.	**B1**

Qns	Answer	Marks
4(b)	e.m.f. is the energy converted from non-electrical to electrical form per unit charge, while p.d. is the energy converted from electrical to non-electrical form per unit charge. [B1]	**B1**
4(c)	As R_2 increases, the total resistance of the circuit increases and the current in the cell decreases. The p.d. across the internal resistance therefore decreases. Hence, using $V_{terminal} = E - Ir$, the terminal p.d. increases. Since R_1 is constant, the current through R_1 increases. -	**M1** **M1** **A1**
4(d)(i)	p.d. across r = 1.40 − (0.220 × 5.50) = 0.19 V. Icell = 0.19/0.13 = 1.46 A ≈ 1.5 A (2 s.f.).	**C1** **A1**
4(d)(ii)	I_2 = 1.5 − 0.220 = 1.28 A; R_2 = (0.220×5.50)/1.28 = 0.945 Ω	**C1** **A1**
	TOTAL FOR QUESTION 4	**9**
5(a)	For a single-slit diffraction pattern, $\sin\theta = \frac{\lambda}{b} \Rightarrow \theta = \sin^{-1}\left(\frac{\lambda}{b}\right) = \sin^{-1}\left(\frac{670\times10^{-9}}{0.12\times10^{-3}}\right) = 0.00558336 \text{ rad (6.s.f)}$ Let the distance between the central maxima to first minima for the diffraction pattern be *y*. Hence, $\tan\theta = \frac{y}{D} \Rightarrow y = D\ (\tan\theta)$ $= 1.20\ (\tan(0.00558336))$ $= 0.00670010 \text{ m (6.s.f)}$ Thus, the width of the central fringe is: 2*y* = 2 (0.00670010) = 0.0134 m (3.s.f) = 13.4 mm (shown) **Note:** Deduct 1 mark if students - used small angle approximation without justification - write the angle in degrees without indicating the degree symbol	**C1** **C1** **M1** **A0**
5(b)(i)	The distance from the central bright maximum to first minimum is: 13.4 / 2 = 6.7 mm. However, the central bright maxima are 2.8 mm away from each other. Hence, we cannot distinguish between the two diffraction patterns from each single slit / they cannot be resolved.	**M1** **A1**

Qns	Answer	Marks
		
5(b)(ii)	$x = \frac{D\lambda}{a} = \frac{(1.2)(670\times10^{-9})}{(2.8\times10^{-3})} = 0.000287143 \text{ m (6.s.f)}$ An estimation of the number of fringes, *n* is $n = \frac{0.0134002}{0.000287143} = 46.66733 = 46$ (round down)	**C1** **A1**
	TOTAL FOR QUESTION 5	**7**
6(a)	Faraday's law of electromagnetism states that the (induced) e.m.f. is directly proportional to the rate of change of magnetic flux linkage.	**B1**
6(b)(i)	$\varepsilon = \frac{d\phi}{dt} = \frac{\Delta\phi}{\Delta t}$ $\Delta\phi = \varepsilon \times \Delta t$ $= 0.55 \times 14 = 7.7$ Wb Accepted units include: Wb / Vs / Tm^2	**A1** **A1**
6(b)(ii)	$\phi = BA$ $A = \frac{\phi}{B} = \frac{7.7}{38\times10^{-6}}$ $= 2.0\times10^{5}$ m^2 **Note:** Allow ECF from **(b)(i)**	**A1**
6(b)(iii)	**Method 1:** Area = speed x time x width $v = \frac{A}{t \times \text{width}}$ $= \frac{2.0\times10^{5}}{14\times68}$ $= 210$ m s^{-1}	**C1** **A1**

Qns	Answer	Marks
	OR **<u>Method 2:</u>** $\varepsilon = \frac{d\phi}{dt}$ $\varepsilon = BLv$ } **C1** $0.55 = \left(38 \times 10^{-6}\right)(68)v$ $v = 213 \text{ m s}^{-1}$ $= 210 \text{ m s}^{-1}$ (2 s.f.) **<u>Note:</u>** Allow ECF from **(b)(ii)**	**OR** **C1** **A1**
6(b)(iv)	As the aircraft travels through the magnetic field with a velocity perpendicular to the field lines, charges in the wing experience a magnetic force. Using Fleming's left-hand rule, magnetic forces act on positive (negative) charges, causing them to move toward wingtip P (Q). As positive (negative) charges accumulate at P (Q) wingtip P will be at a higher potential than Q. sp	**B1** **A1**
6(c)(i)	<table><tr><th>Resistor</th><th>r.m.s. voltage, $V_{r.m.s.}$ / V</th><th>peak voltage, V_0 / V</th></tr><tr><td>**R_1**</td><td>$\frac{16}{\sqrt{2}}$ = 11 (or 11.3)</td><td>16</td></tr><tr><td>**R_2**</td><td>$\frac{16}{2}$ = 8.0</td><td>16</td></tr></table>	**A1**
6(c)(ii)	The peak voltages are the same because when the diode is forward-biased, its <u>resistance is close to zero</u>, giving both resistors the same peak potential difference. The r.m.s. voltages for both resistors are different because the branch with resistor R_2 and the diode undergoes <u>half-wave rectification</u>, unlike the branch with only resistor R_1.	**B1** **B1**
	TOTAL FOR QUESTION 6	**11**
7(a)	Half-life is the <u>average</u> time taken for half the original <u>number</u> of radioactive <u>nuclei</u> to decay.	**B1**
7(b)(i)	$A = \lambda N$ $1.6 \times 10^{14} = \frac{ln\,2}{3.6 \times 24 \times 3600} N$ [M1] $N = 7.18 \times 10^{19}$ Mass = $7.18 \times 10^{19} \times 224 \times 1.66 \times 10^{-27}$ [M1] $= 2.67 \times 10^{-5}$ kg $= 27$ mg [A0]	**M1** **M1** **A0**

8(a)(i)	$T = RC\ln\left(\frac{\varepsilon - V_s}{\varepsilon - V_b}\right)$ $= 18\times10^{3}\times33\times10^{-6}\ln\left(\frac{5.4-0.9}{5.4-4.5}\right)$ [C1] $= 0.956\text{s}$ $f = \frac{1}{T} = 1.046\,\text{s}^{-1}$ = 1.05 Hz [M1] Comment: Passage mentioned: steady heart rate is 60 beats per minute, the frequency of 1.046 s^{-1} is about 62 beats per min which is close to steady heart rate. Hence it is appropriate. [A1] *Candidate must compare current frequency with steady heart rate from passage to gain this mark.* Award [B1] for any one physiologically valid situation where a fixed pacing rate is inappropriate, with a brief explanation of why. Examples: **Physical activity**: cannot increase rate during exercise leading to insufficient cardiac output during exertion **Sleep**: healthy hearts slow to 40–50 bpm during deep sleep hence fixed rate may be unnecessarily high, consuming more battery **Emotional stress/anxiety**: heart rate naturally rises and fixed rate cannot respond **Illness/fever**: body may require higher heart rate and fixed rate cannot adapt **Individual variation**: 60 bpm may not be the optimal resting rate for every patient but there is no way to adjust without surgery to replace resistor **Competing with intrinsic rhythm**: if patient's own heart occasionally fires, fixed-rate pacemaker may deliver a pulse at the wrong moment in the cardiac cycle, potentially dangerous p.s. this is actually the most clinically serious one, known as "R-on-T phenomenon"	**C1** **M1** **A1** **B1**
8(a)(ii)	τ_charging = RC = $18\times10^{3}\times33\times10^{-6}$ = 0.594 s □ τ_discharge = R_L C = $500\times33\times10^{-6}$ = 0.0165 s [M1 for both calculations] □ τ_discharge is 36× smaller than τ_charging (or compare and say about 2% only) ; since discharge time << charging time << T, the assumption that discharge is almost instantaneous is well justified. [A1]	**M1** **A1**
8(b)(i)	The energy *U* stored in a capacitor of capacitance *C* charged to potential difference *V* is: $U = \frac{1}{2}CV^2$ Since the capacitor charges to V_b before each discharge and retains charge at *Vs* after each discharge, the energy ΔU delivered to the heart per pulse is: $\Delta U = \frac{1}{2}C\left(V_b^{\,2} - V_s^{\,2}\right)$ = ½ (33 x 10^{-6})($4.5^2 - 0.9^2$) = 3.21 x 10^{-4} J	**C1** **M1** for correct working + substitution A0
8(b)(ii)	Power = energy / time = energy x frequency	

	P = 3.21 x 10^{-4} x 1.046 = 0.0003357 = 3.36 x 10^{-4} W	M1 A1
8(b)(iii)	Each pulse is said to be almost "instantaneous" which means the power is delivered only during the discharging process [B1] and during the charging process, no energy is delivered to the heart tissue [B1]; hence the time-averaged power is much lower than the power during each pulse	B1 B1
8(c)(i)	Converts capacity to coulombs: Q = 1.2 × 3600 = 4320 C **[C1]** Energy = VQ = 5.4 × 4320 = 23328 J = 23.3 kJ (3sf) **[A1]**	C1 A1
8(c)(ii)	Lifespan = E_batt / P_avg = 23328 / (3.36×10^{-4}) = 6.94×10^{7} s = 2.2 years Very short lifespan, will require operation every 2.2 years to change battery. OR 2.2 years is an overestimate because the calculation assumes all stored battery energy is ultimately delivered to the heart and neglects losses/other circuit consumption.	M1 A1
8(d)(i)	Increase in C → increase in T calculated from (a)(i)/ time constant → longer charging time and longer period T → lower pacing frequency since f = 1/T. [1]	B1
8(d)(ii)	A patient with complete heart block is entirely dependent on the pacemaker for ventricular stimulation; if the pacing rate drops, their heart rate drops correspondingly. [B1] If the pacing rate falls below ~40 bpm (as stated in passage), the patient risks fainting / cardiac arrest. [B1] A slow gradual drift may go unnoticed without regular monitoring, person's health may deteriorate slowly, making it potentially life-threatening / dangerous. [B1] Award max 2 points for any 2 points.	B1 B1
	TOTAL FOR QUESTION 8	**19**

Qns	Answer	Marks
(b)(ii)	Binding energy = $1.767 \times 1.66 \times 10^{-27} \times (3.00 \times 10^{8})^2$ $= 2.64 \times 10^{-10}$ J Binding energy per nucleon = $[2.64 \times 10^{-10} / 1.60 \times 10^{-19} \times 10^{6}]/210$ = 7.86 MeV	M1 A1

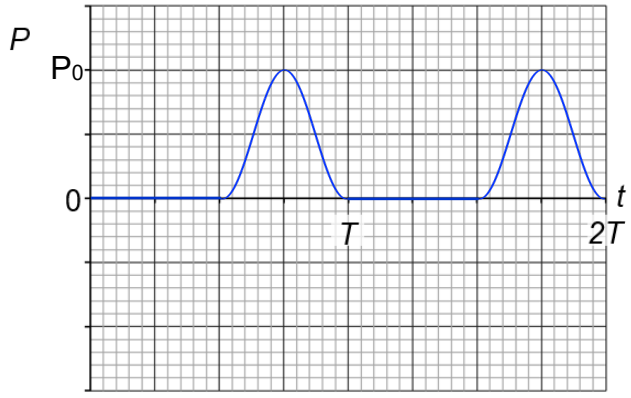
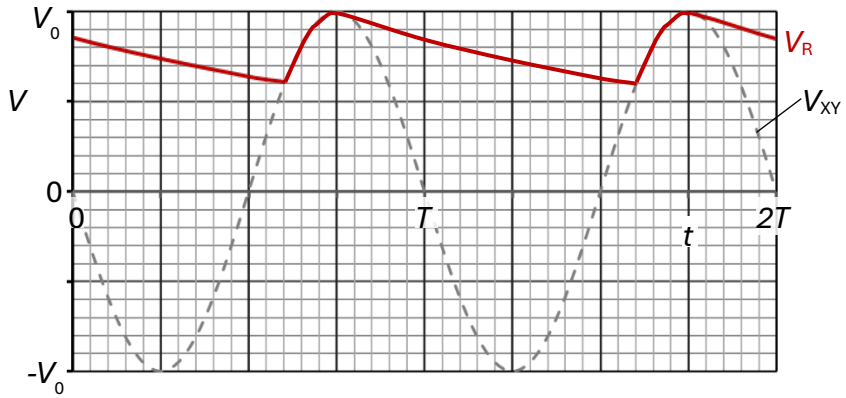
Qns	Answer	Marks
8(a)1	Since the circular path is in the clockwise direction it suggests that the magnetic force must point rightwards (from left to right) Using Fleming's Left Hand Rule, (when the magnetic field is out of the paper, and the magnetic force is towards the right), this is only possible if the particle is positively charged.	B1 B1
8(a)2	For the positively charged particle to accelerate upwards it must experience an electric force that acts in the upward direction. This occurs only if the electric field points upwards which corresponds to a higher electric potential at the bottom plate.	B1 B1
8(b)(i)	The electric field between parallel plates is uniform. $E = \frac{\Delta V}{\Delta d} = \frac{V_0}{d} = \frac{V}{0.5d}$ Hence, $V = \frac{V_0}{2}$	M1 A0
8(b)(ii)	Gain in kinetic energy = Loss in electric potential energy $\frac{1}{2}mv^2 - 0 = q\Delta V = \frac{qV_0}{2}$ $\therefore v^2 = \frac{qV_0}{m}$ $v = \sqrt{\frac{qV_0}{m}}$	M1 A0
8(c)	The magnetic force always acts perpendicular to the direction of velocity (infinitesimal tangential displacement along the charged particle's path). No work is done, its kinetic energy remains constant and thus, the speed does not increase.	B1 B1

Qns	Answer	Marks
8(d)	The magnetic force (F_B) provides for the centripetal force (F_c) $F_B = F_c$ $Bqv = mr\omega^2 \Rightarrow Bq(r\omega) = mr\omega^2$ $\therefore \omega = \frac{Bq}{m}$ $\omega = \frac{2\pi}{T}$ $\frac{2\pi}{T} = \frac{Bq}{m}$ $\therefore T = \frac{2\pi m}{Bq}$ Since the particle travels half a circle, the time travelled in the upper dee is: $t = \frac{T}{2}$ $= \frac{\pi m}{Bq}$ **Alternative solution:** The magnetic force (F_B) provides for the centripetal force (F_c) $F_B = F_c$ $Bqv = \frac{mv^2}{r} \Rightarrow v = \frac{Bpr}{m}$ Since $v = \frac{\pi r}{t}$, where πr is the distance travelled and t is the time taken for the semicircular path $\frac{\pi r}{t} = \frac{Bqr}{m}$ $\therefore t = \frac{\pi m}{Bq}$	B1 C1 M1 M1 A0 B1 C1 M1 M1 A0
8(e)	The time taken is independent of the velocity of the particle (only depends on m, B and q), Hence, it will always be the same as that found in (d).	M1 A1
8(f)	t = 1.6 ns $B = \frac{\pi m}{tq} = \frac{\pi}{(1.6\times10^{-9})(1.0\times10^{8})} = 19.63495 = 20$ T (2.s.f)	C1 A1
8(g)	At the upper dee, the charged particle has a kinetic energy of $\frac{qV_0}{2}$ Gain in kinetic energy = Loss in electric potential energy = qV_0	 C1

Qns	Answer	Marks
	Thus, total kinetic energy = $\frac{qV_0}{2} + qV_0 = 1.5qV_0$	**A1**
8(h)	When the particle first travels into the upper dee, it has a kinetic energy of 0.5 qV_0. The total kinetic energy of the particle can then be written as E_K = (0.5 + n) qV_0, where n is the number of times the particle is in either of the dees after exiting the upper dee for the first time. Since the kinetic energy is 6.5 qV_0, n = 6. By doing a manual count, the particle will exit from the upper dee and the it passes the lower dee is a total of 3 times. **#Food for thought:** To generalize this: If n = odd, then we can tell that the particle exits from the lower dee and the number of times it will pass the lower dee is $\frac{(n+1)}{2}$ If n = even, then we can tell that the particle exits from the upper dee and the number of times it will pass the lower dee is $\frac{n}{2}$	**C1** **A1**

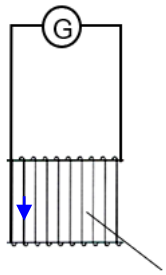
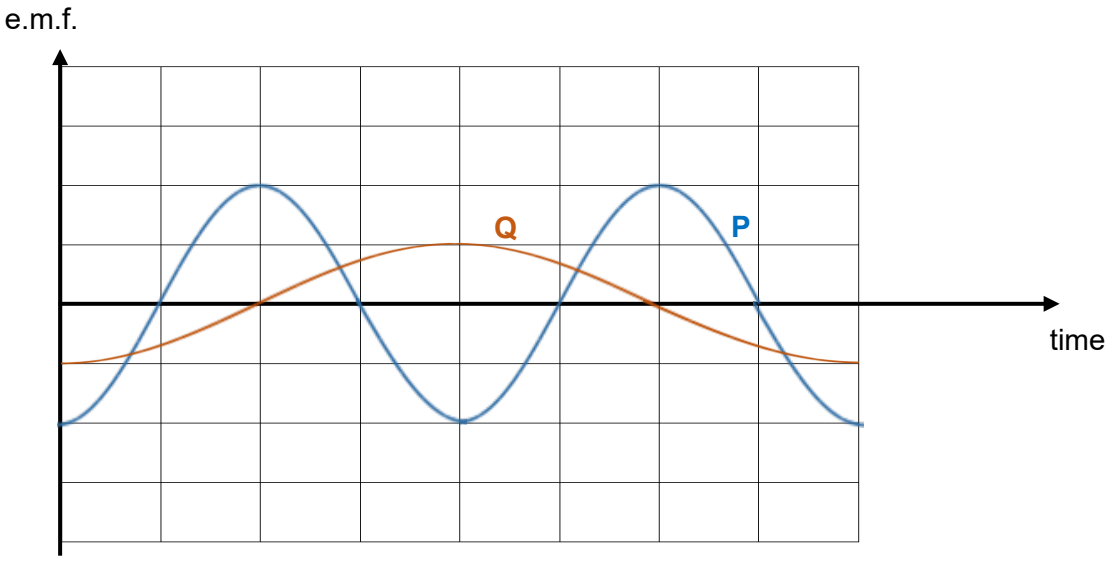
Qns	Answer	Marks
9(a)(i)	The square of the wavefunction amplitude $\lvert\psi(x)\rvert^2$ is the probability density function of the particle at position x.	**B1**
9(a)(ii)	The probability of the particle residing between x_1 and x_2 is given by the area under the $\lvert\psi(x)\rvert^2$-x graph between x_1 and x_2.	**B1**
9(b)(i)	n = 4.	**B1**

Question 5		
(a)	Electric field between plates M and N $E = \frac{V}{d} = \frac{235-(-235)}{2.00\times10^{-2}} = 2.35\times10^{4}\ \text{V m}^{-1}$ $F = Eq = \left(2.35\times10^{4}\right)\left(58\times10^{-12}\right) = 1.4\times10^{-6}\ \text{N}$	**B1** **A1**
(b)	Acceleration, $a = \frac{F}{m} = \frac{1.363\times10^{-6}}{0.12\times10^{-3}} = 1.136\times10^{-2} = 1.1\times10^{-2}\ \text{m s}^{-2}$	**A1**
(c)	Time to travel from M to N $s = ut + \frac{1}{2}at^2$ $2.0\times10^{-2} = 0 + \frac{1}{2}\left(1.136\times10^{-2}\right)\left(t^2\right)$ $t = 1.8765\ s$ Frequency, $f = \frac{1}{2\times t} = \frac{1}{2\times1.8765} = 0.266\ \text{Hz}$	**B1** **B1** **A1**
(d)	The acceleration for simple harmonic motion is proportional to its displacement. Since the acceleration of the sphere is constant, the motion cannot be simple harmonic.	**A1**

Question 6		
(a)(i)	Two sections of zero power of width $0.5T$ starting at $t = 0$ and $t = T$ Two humps of width $0.5T$ and peak P_0 *starting* at $t = 0.5T$ and $t = 1.5T$ [Deduct only this mark if the positions of the zero power and humps are swapped.] Correct sinusoidal-squared shape for the 2 non-zero humps (i.e. with smooth troughs seating on the t-axis at $P = 0$ 	**B1** **B1**
(a)(ii)	The mean power within each hump is $\frac{1}{2}P_0$ from the symmetry of the curve. An additional half factor is from the removal of half of the power in each cycle (resulting in the ¼ factor).	**B1** **B1**
(a)(iii)	$\frac{V_{OUT,rms}^{\ 2}}{R} = \frac{1}{4}P_0 = \frac{1}{4}(2\frac{V_{XY,rms}^2}{R})$ $V_{OUT,rms} = \frac{V_{XY,rms}}{\sqrt{2}} = \frac{12}{\sqrt{2}} = 8.485 = 8.5\ \text{V}$	**M1** **A1**
(b)(i)	Correct shape.  Note: No penalty if the first part of the graph is not drawn.	**B1**
(b)(ii)	From the graph, the time for the capacitor to discharge from V_0 to 0.60 V_0 is $0.85T$. $V = V_0 e^{-\frac{t}{\tau}} \Rightarrow \tau = \frac{-0.85T}{\ln 0.60} = 1.7T$	**B1** **A1**

Question 5		
(a)	$I = \frac{E}{R} \Rightarrow \frac{\Delta I}{I} = \frac{\Delta R}{R}$ (E is a constant.) $\Delta I = 5\% \times (\frac{E}{R}) = 0.05 \times \frac{9.0\text{ V}}{10\ \Omega} = 0.045$ A = 0.05 A (to 1 s.f.) <u>Alternative:</u> Max. value of R = 1.05 x 10 Variation in I = $\frac{9.0}{10} - \frac{9.0}{1.05 \times 10} = 0.0428$ A ~ 0.05 A <u>Alternative:</u> • min value of R - variation will be 0.0474 A, or • (max – min) /2 (variation will be 0.0451 A)	**B1** **A1**
(b)(i)	Any correct explanation, e.g.: • The actual results differs from the predicted results by more than 5% (up to approximately 40%). • If it was a random variation, then the actual results should deviate in such a way that it sometimes higher and sometimes lower than the predicted results (but still producing a straight line trend). • Actual results is always lower than the predicted results suggesting a systematic error is present.	**A1**
(b)(ii)	As more resistance is added to the parallel network, the **effective resistance of the network of resistors reduces**. The external circuit and the internal resistance of the battery form **a potential divider**, so a **smaller fraction of the emf is across the external circuit / the terminal p.d. decreases**, causing the measured current to be increasingly lower than the predicted current. <u>Alternative:</u> As more resistors are connected in parallel, **the effective resistance decreases and the current drawn increases.** Therefore, the **potential drop across the internal resistance increases**, so the **terminal p.d. across the external circuit decreases** and the measured current becomes increasingly lower than predicted.	**B1** **B1** **B1** **B1**
(c)	$E = IR + Ir \Rightarrow \frac{1}{I} = (\frac{1}{E})R + \frac{r}{E}$ $\Rightarrow \frac{r}{E} = r$(gradient) = vertical intercept Using (1.6, 0.28) and (9.0, 1.12), Gradient of graph = $\frac{(1.12) - (0.28)}{(9.0) - (1.6)} = 0.1135$	**B1** **B1**

	Note to marker: • Accept anything from 0.10 to 0.12. • Deduct this mark if working is not shown for gradient. $r = \frac{\text{vertical intercept}}{\text{gradient}} = \frac{(0.10)}{0.1135} = 0.88\ \Omega$ Accept solution if student substitutes point(s) from best-fit line and solve simultaneous equations to get the final internal resistance e.g. using two points on BFL or a point with gradient. [-1] if the student uses data point does not lie on the BFL or use 9.0 V as the emf.	**A1**

Question 6		
(a)(i)	The e.m.f. induced in a conductor is proportional to the rate of change of magnetic flux linkage. Magnetic flux density generated by solenoid is decreasing (as the current flowing through the solenoid is decreasing). Hence, magnetic flux linkage in small coil is decreasing, so e.m.f. is induced.	**B1** **B1**
(a)(ii)	By Lenz's law, the magnetic field set up in the small coil due to the induced current will be in **the same (rightward) direction as the magnetic field generated by the solenoid** to **oppose the decrease in the flux linkage.** Apply right hand grip rule to determine direction of induced current. Direction of induced current marked correctly on diagram (Induced current should be drawn on a wire and pointing down).  	**B1** **A1**
(b)(i)	 Cosine graph plotted (positive or negative accepted) $I = I_0 \sin\omega t + c$ where $c > 0$ Since $B \propto I$, $\varepsilon = -\frac{d(NBA)}{dt} = -NA\frac{dB}{dt}$ Hence, $\varepsilon \propto -\cos\omega t$	**A1**
(b)(ii)	Peak value halved Period doubled	**A1** **A1**

4 **(a)** Fig. 4.1 shows a circular coil of 500 turns and radius 0.12 m.

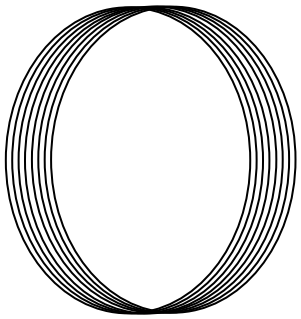


Fig. 4.1

A varying magnetic field is applied at right angles to the plane of the coil. Fig. 4.2 shows the variation with time *t* of this magnetic field of flux density *B*.

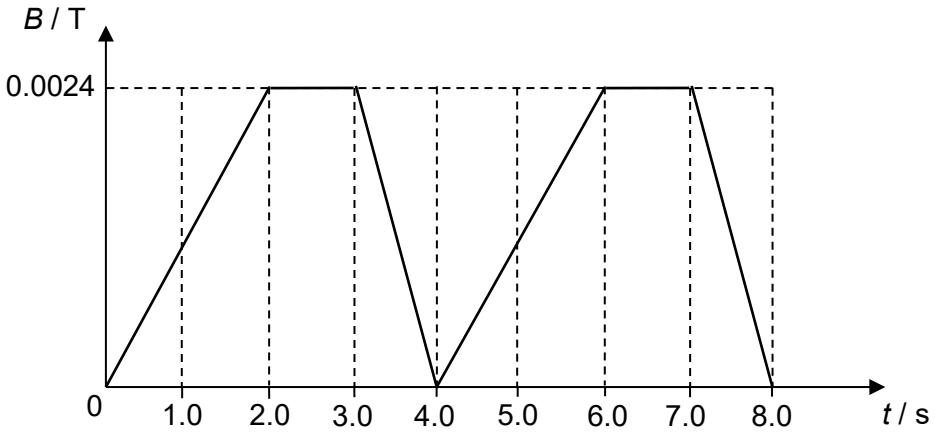


Fig. 4.2

(i) On Fig. 4.3, sketch a graph to show the variation with time t of the induced e.m.f. *E* in the coil. Include values on the y axis.

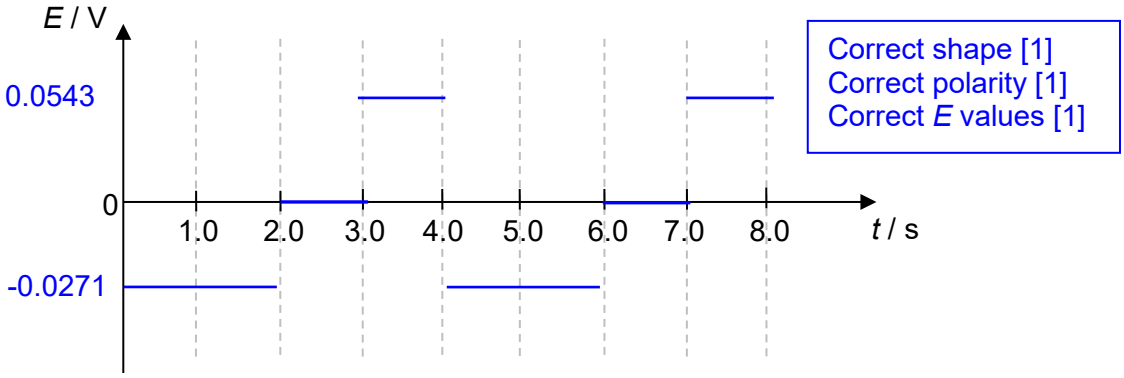


Fig. 4.3

[3]

(ii) Hence, determine the r.m.s value of the induced e.m.f.

$$\text{r.m.s.} = \sqrt{\frac{(0.0271^2 \times 2) + (0.0543^2 \times 1)}{4}} \quad \text{M1}$$

$$= 0.0332\text{A} \quad \text{A1}$$

r.m.s. value of induced e.m.f. = ... V [2]

(iii) The e.m.f. source that you have sketched in Fig. 4.3 is now connected in series with a resistor *R* and an ideal diode as shown in Fig. 4.4.

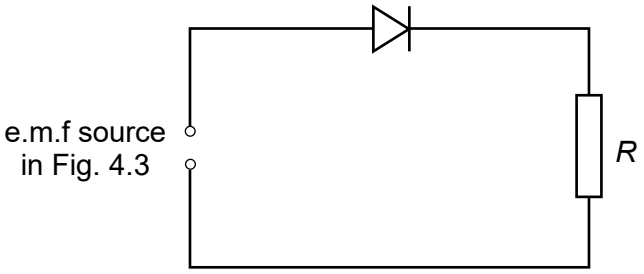


Fig. 4.4

State and explain the potential difference across R when the ideal diode has maximum potential difference across it.

When diode has maximum p.d. across it, it is reverse bias / not conducting / has infinite resistance. [M1].

The sum of the p.d. across R and diode must equal to the e.m.f. of the supply. [M1]

Since resistance of R is insignificant compared to that of diode, p.d. across R is zero [A1]. OR The total resistance of the circuit is infinite and the current in the circuit is zero. [M1] Since there is no current in R, p.d. across R is zero. [A1]

[3]

[Total: 8]

5 **(a)** A straight rigid conducting wire of mass 0.14 kg and length 0.40 m is placed horizontally on a frictionless plastic plane. The plane is at an angle of 20° to the horizontal and is placed in a region of upwards uniform vertical magnetic field as shown in Fig. 5.1 and Fig. 5.2.

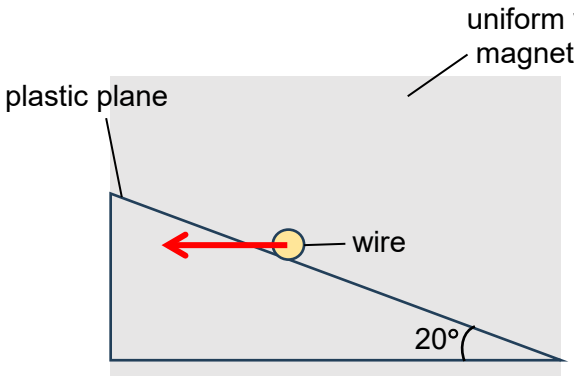


Fig. 5.1 (side view)

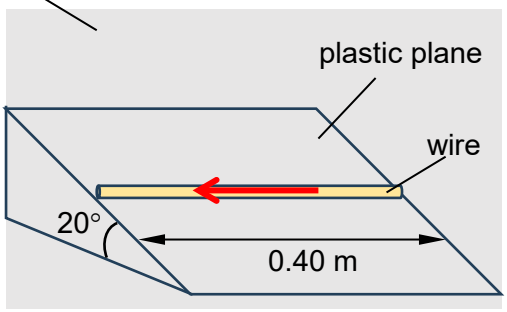


Fig. 5.2

To prevent the wire from sliding down the plane, a steady current I is passed through the wire so that it remains stationary on the plastic plane.

(i) On Fig. 5.1, draw an arrow to show the direction of the magnetic force acting on the wire. [1]

(ii) On Fig. 5.2, draw an arrow to show the direction of the current I in the wire. [1]

(iii) With reference to forces acting on the wire, explain why the wire remains stationary.

The direction of current I is perpendicular to the magnetic field/ flux density B and by Flemings left hand rule, experiences a force perpendicular to both I and B, acting leftwards.(Fig. 5.1)

[B1] The wire remains stationary when the component of the magnetic force upwards along the plane balances the component of weight downwards along the plane. [2]

[2]

(iv) Suggest and explain whether the wire will eventually slide up, remain stationary or slide down the plane if an alternating current of root-mean-square value of I is passed through the wire instead.

The current I will alternate between positive and negative values, and the component of magnetic force ($\propto I$) will alternate between acting upwards and downwards along the plane.

Since the average I and hence magnetic force between cycles is zero, it cannot balance the component of weight downwards along the plane. Hence the rod will slide down the plane.

Or?

The current I will alternate between positive and negative values, and the component of magnetic force ($\propto I$) will alternate between acting upwards and downwards along the plane. [2]

Since I_{rms}= 0.707 I_0, the time duration at which the a.c current causes a **net** upward force along plane (when upward $F_{m//}$ > W// to slope downwards) is shorter than when there is a net downward force. Hence the rod will slide down the plane.

(b) The current in **(a)** is disconnected, and the magnetic field is removed. The wire is now given a uniform net positive charge of $+2.0 \times 10^{-6}$ C.

(i) To keep the wire stationary in the same position on the frictionless plane, a uniform electric field of strength E is applied in the horizontal direction.

Determine the magnitude of E.

The electric force must be acting leftwards for wire to be stationary. $\sum F_{//} = 0$

$$F_{E//} = W_{//}$$

$$qE_{//}\cos\theta = mg\sin\theta \quad \text{[C1]}$$

$$E_{//} = \frac{mg}{q}\tan\theta$$

$$= \frac{(0.14)(9.81)}{2.0\times10^{-6}}\tan 20° \quad \text{[M1]}$$

$$= 2.5\times10^{5} \quad \text{[A1]}$$

E = ... N C^{-1} [3]

[Total: 9]

9 **(a)** In a circuit, a 4.0 V battery with internal resistance is connected in series to a thermistor and a coaxial resistor.

The coaxial resistor has a cylindrical core of length 0.020 m. The core is made of material A and is surrounded by material B as shown in Fig. 9.1. In the circuit, the direction of the current passing through the coaxial resistor is parallel to its length. The physical dimensions of the coaxial resistor are fixed.

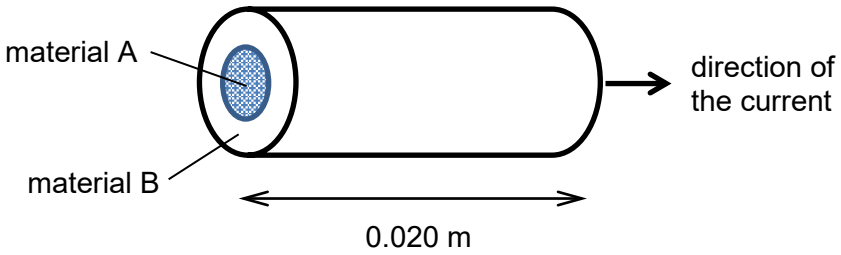


Fig. 9.1

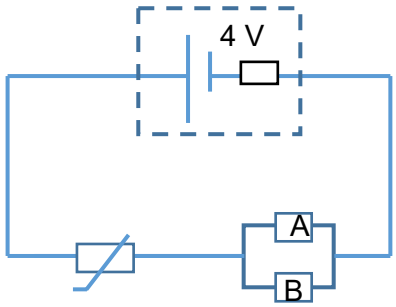
(i) Draw a complete circuit diagram of the circuit in the space below.

Represent the coaxial resistor as two separate resistors A and B, one for each material respectively.

thermistor B1

parallel B1

internal resistance B1



[3]

(ii) Fig. 9.2 shows the variation of the current I in the thermistor with temperature T of the thermistor.

Fig 9.3 shows the variation of I in the thermistor with voltage V across the thermistor.

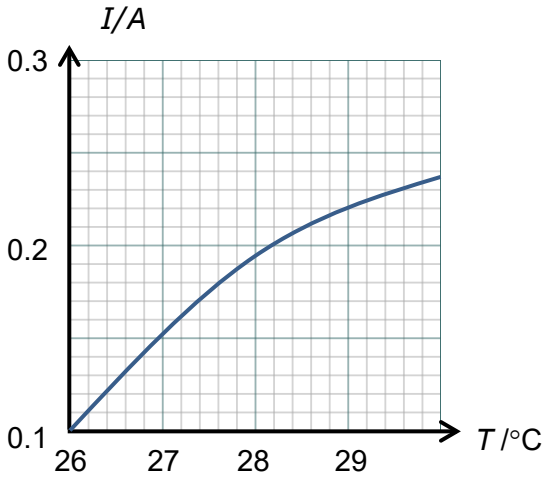


Fig. 9.2

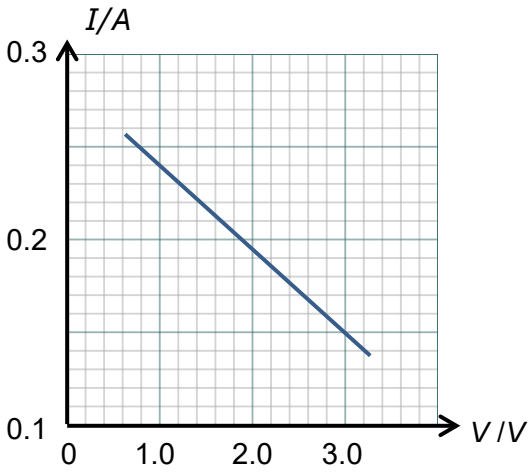


Fig. 9.3

Determine the resistance of the thermistor at 27 °C.

Reading from the graph, at 27°C, I = 0.15 A

And pd across resistor = 3 V C1

Hence resistance of thermistor = $\frac{3}{0.15} = 20\ \Omega$ A1

resistance = ... Ω [2]

(iii) Describe and explain what will happen to the potential difference across resistors A and B in the circuit when the temperature of the thermistor increases.

By potential divider rule, $V_A = \frac{R_{effA,B}}{\text{Total resistance}} \times E$ M1

Since total resistance drop, V_A and hence V_B increases since they are in parallel. A1

[2]

(b) (i) Two identical lamps are connected to a d.c. and a.c. supply as shown in Fig. 9.4a and Fig. 9.4b respectively.

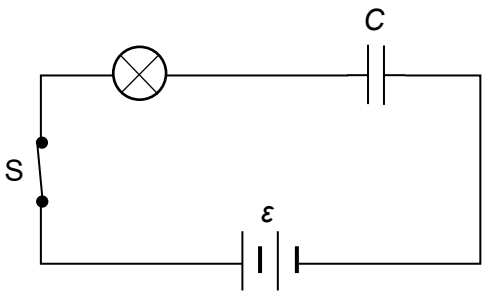


Fig 9.4a

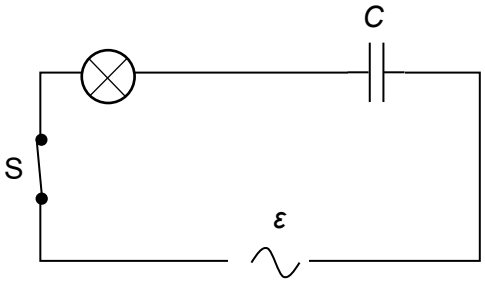


Fig 9.4b

State and explain which lamp will be turned on for a longer period of time. Assume that the time constant is large.

In 9.4a, the lamp will be on when the capacitor is charging. At full charge, the lamp will be off B1

In Fig 9.4b, when the capacitor is constantly charged and discharged because of the periodic nature of the a.c.

Hence a continuous supply of ac passes through the lamp to light it and it will stay lit for a longer period of time.

[3]

(ii) The charged capacitor of capacitance C is being discharged through a resistor of resistance R, as shown in Fig. 9.5. At time = 0 s, the capacitor is fully charged.

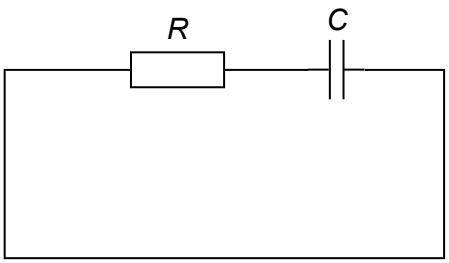


Fig. 9.5

The energy stored in the capacitor decreases with time as the capacitor discharges. Determine the time t, in terms of R and C, it takes for this stored energy to be discharged by 75% of its initial value.

Energy = $\frac{q^2}{2C}$

= $\frac{Q^2}{2C}e^{-\frac{2t}{RC}}$ B1 (if took as 3/4 can get method mark for this)

= $U_0 e^{-\frac{2t}{RC}}$

Hence $\frac{1}{4}U_o = U_0 e^{-\frac{2t}{RC}}$ M1

$t = \frac{1}{2}RC\ln 4 = 0.693RC$ A1

t = ... [3]

(c) After a long time, the capacitor in Fig. 9.5 is fully discharged and a battery of e.m.f E of negligible internal resistance is used to recharge the capacitor in a circuit as shown in Fig. 9.6.

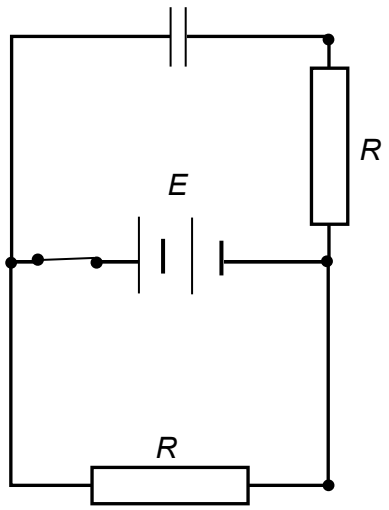


Fig. 9.6

(i) Determine the current, in terms of E and R, in the battery immediately after the switch is closed.

$R_{eff} = \frac{R}{2}$ M1

$I = \frac{E}{2R}$ A1

current = ... [2]

(ii) Determine the current, in terms of E and R, in the battery a long time after the switch is closed.

When the capacitor is fully charged, there is no current in the resistor in series with it. M1

Hence $I = \frac{E}{R}$ A1

current = ... [2]

(iii) Hence suggest an expression for the time constant, in terms of R and C.

Explain your reasoning.

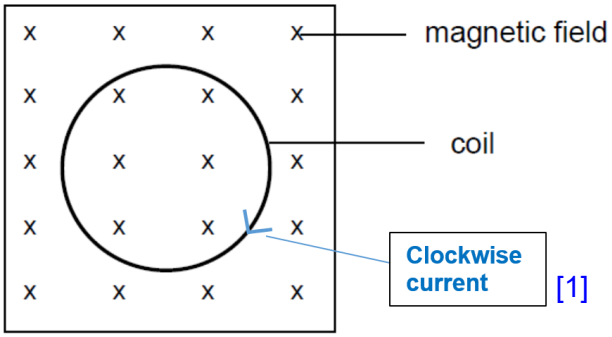
The capacitor and its resistor (the one in series with it) are parallel to the lower resistor. B1 Hence, the pd across the lower resistor is always E and thus (will not affect) can be removed without affecting the charging process. M1 Hence, only the resistor in series with it will affect the charging process and thus the time constant is RC. A1

[3]

[Total: 20]

END OF PAPER

5 A coil is placed in a region of magnetic field which is directed into the plane of the paper as shown in Fig. 5.1.



[1]

Fig. 5.1

(a) Define *magnetic flux*.

The magnetic flux through a surface is the product of the component of the magnetic flux density normal to the surface and the area of the surface [1].

[1]

(b) **(i)** The magnitude of the flux density of the magnetic field is decreasing with respect to time. Explain, using Faraday's law, why there is an induced *current* in the coil.

Since the flux density B decreases, then flux linkage Φ = BA decreases. [1]

By Faraday's Law, there is an induced emf in the coil. Since the coil is a closed circuit, the induced emf produces an induced current. [1] [2]

(ii) Draw, on Fig. 5.1, the direction of the induced current. Explain, using Lenz's law, why the induced current is in this direction.

By Lenz's Law, the direction of the induced current must be such as to produce a magnetic flux into the page to *oppose* the decrease of the magnetic flux linkage. [1] Therefore, by the Right Hand Grip Rule, the current will be in the *clockwise* direction to produce an induced magnetic field in the same direction. [2]

[1 for correct direction of induced current]

(c) A magnet of length *x* and magnetic flux density *B* is dropped vertically from rest with the north pole facing downwards. It is dropped from a reference line which is at a vertical height *h* above the top of a *long* solenoid of length *L* and uniform cross-sectional area as shown in Fig. 5.2. The length of the magnet *x* is much smaller than the length of the solenoid *L*.

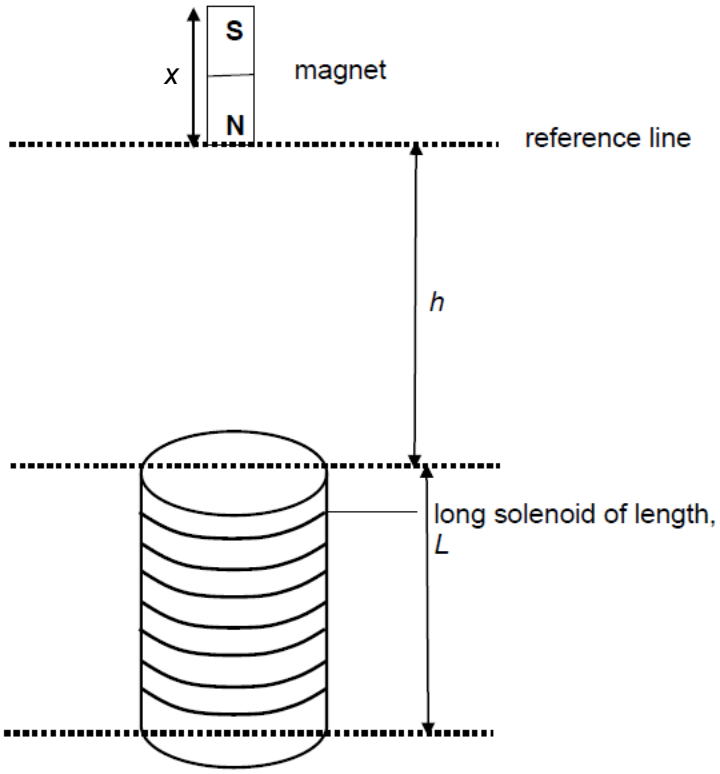


Fig. 5.2

(i) At time t = 0.0 s, the magnet is released from the reference line and allowed to pass through the entire length of the solenoid. Sketch, on Fig. 5.3, a graph to show the variation of the induced e.m.f E in the solenoid with time t. Label this graph as A. (Numerical values are not required and ignore the effects of air resistance.)

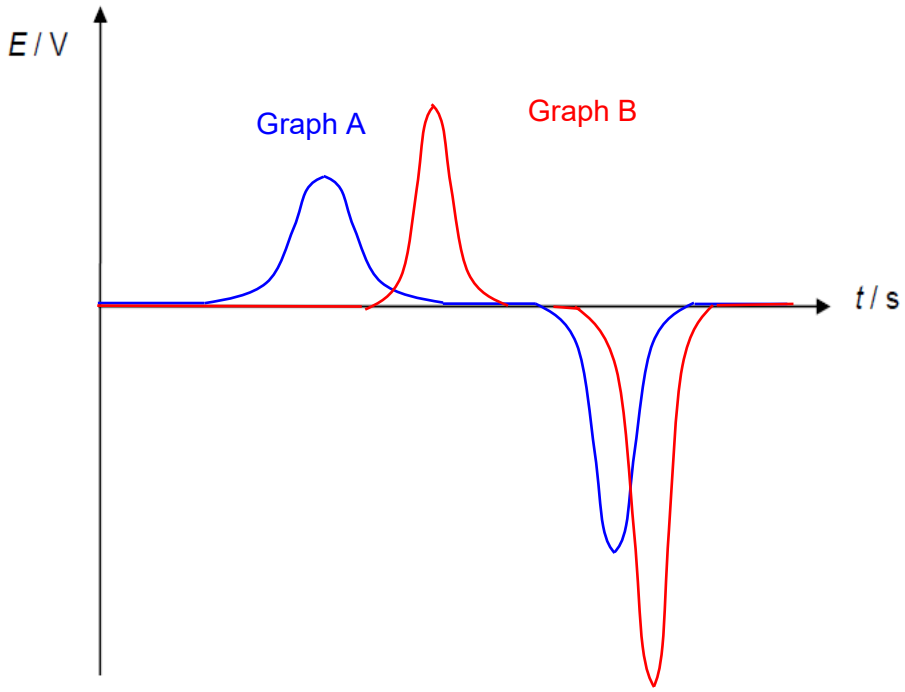


Fig. 5.3

[2]

(ii) State how your graph in part **(c)(i)** would change if the same magnet is released with the south pole facing downwards instead.

The new graph will look like a reflection of Graph A about the time axis [1].

[1]

(iii) The experiment is now repeated with the north pole of the magnet facing downwards, but released from a greater height above the solenoid. Sketch, on Fig. 5.3, a new graph to show the variation of the induced emf E in the solenoid with time t. Label this graph as B.

Both peaks are higher and narrower [1]

Both peaks occur at later time [1]

[2]

[Total: 10]

4 **(a)** Define *electric field strength*.

Electric field strength at a point is defined as the **electric force per unit positive charge** exerted on a **small test charge** placed at that point.

.....[1]

(b) An oil drop of mass 1.0×10^{-15} kg and carrying a charge of 1.6×10^{-19} C is placed in the middle of two horizontal parallel plates.

The separation between the plates is 40 mm and the the potential difference between the plates is 8.8 kV. The oil drop is seen to be attracted towards the upper plate because of its electric charge.

(i) Calculate the electric force experience by the oil drop.

Electric force = qE

$$= q\frac{V}{d} \quad [1]$$

$$= \left(1.6\times10^{-19}\right)\frac{8.8\times10^{3}}{40\times10^{-3}}$$

$$= 3.52\times10^{-14} \text{ N} \quad [1]$$

force = ... N [2]

(ii) Initially the oil drop is 20 mm from either plate and is at rest. Ignoring air resistance, use energy considerations to calculate its speed when it reaches the upper plate.

By Conservation of Energy,
Gain in kinetic energy + Gain in GPE = Loss in electrical potential energy

$$\frac{1}{2}mv^2 - 0 = q\Delta V + mg\Delta h \quad [1]$$

$$\frac{1}{2}\left(1.0\times10^{-15}\right)v^2 = \left(1.6\times10^{-19}\right)\left(\frac{8.8\times10^{3}}{2}\right) + \left(1.0\times10^{-15}\right)\times9.81\times(20\times10^{-3})$$

$$v = 1.01 \text{ m s}^{-1} \quad [1]$$

speed = ... m s^{-1} [2]

[Total: 5]

5 During cosmic-ray experiments in the 1930s, researchers observed the track of a previously unknown positively charged particle in a cloud chamber. The particle was measured to have a charge of +1 *e* and is known as the muon.

The particle enters a uniform magnetic field of flux density 0.250 T with a speed of 2.40×10^{7} m s^{-1}. Its velocity makes an angle of 60° with the direction of the magnetic field, causing it to follow a helical path as seen in Fig. 5.1 below.

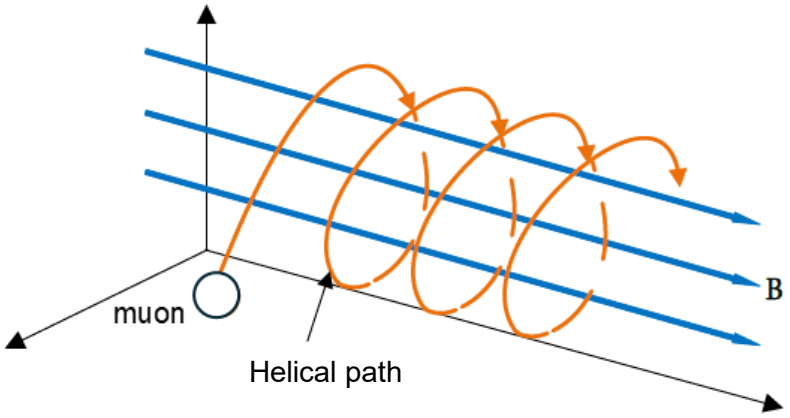


Fig. 5.1

Measurements of the track show that the muon took 2.96×10^{-8} s to complete one revolution.

(a) (i) Show that the radius of the helical path is 0.0979 m.

$$v_{\perp} = r\omega = r\frac{2\pi}{T}$$

$$r = \frac{v_{\perp}T}{2\pi}$$

$$= \frac{2.40\times10^{7}\sin60^{\circ}\times2.96\times10^{-8}}{2\pi}$$

$$= 0.09792$$

$$\approx 0.979 \text{ m}$$

[2]

(ii) Determine the mass, m_{μ} of the muon.

The magnetic force provides the centripetal force:

$$Bqv_{\perp} = \frac{mv_{\perp}^{2}}{r}$$ [1]

$$m = \frac{Bqr}{v_{\perp}}$$

$$= \frac{0.250\times1.60\times10^{-19}\times0.0979}{2.40\times10^{7}\sin60^{\circ}}$$

$$= 1.88\times10^{-28} \text{ m}$$ [1]

m_{μ} = ... kg [2]

(b) Determine the distance travelled by the particle **parallel** to the magnetic field during one complete revolution.

Distance moved in 1 revolution

$d = v_{\parallel} T$

$= v \cos 60^{\circ} T$ [1 for correct resolution]

$= 2.4 \times 10^{7} \times \cos 60^{\circ} \times 2.96 \times 10^{-8}$

$= 0.355$ m [1]

distance = ... m [2]

(c) Before taking measurements, the researchers compensate for the vertical component of the Earth's magnetic field.

A short magnetic needle is suspended from a string 5.0 cm from a horizontal long straight wire, as shown in the top view in Fig. 5.2.

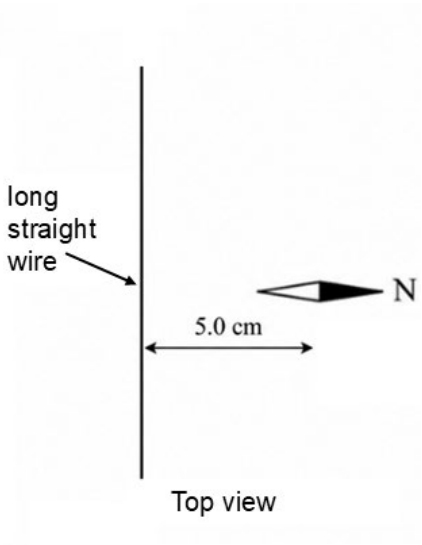


Fig. 5.2

When there is no current in the wire, the north pole of the needle dips at an angle of 6.0° below the horizontal, as shown in the side view in Fig. 5.3.

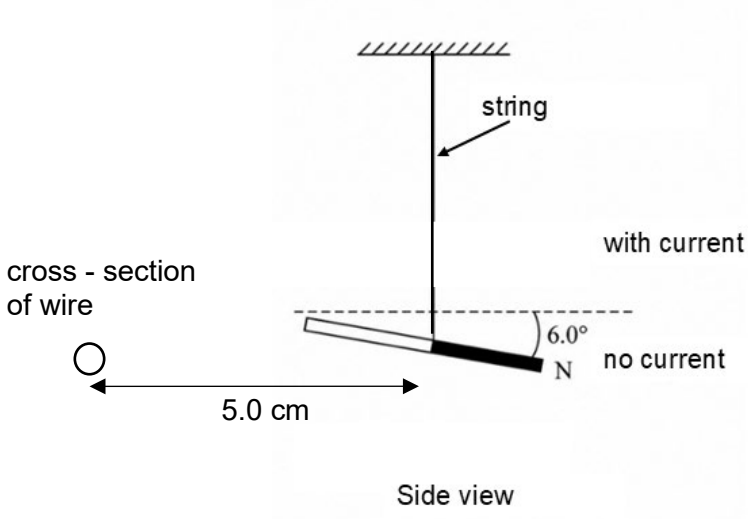


Fig. 5.3

A current of 0.48 A is then passed through the wire. The magnetic field produced by the current causes the needle to become horizontal, as shown in the side view in Fig. 5.4.

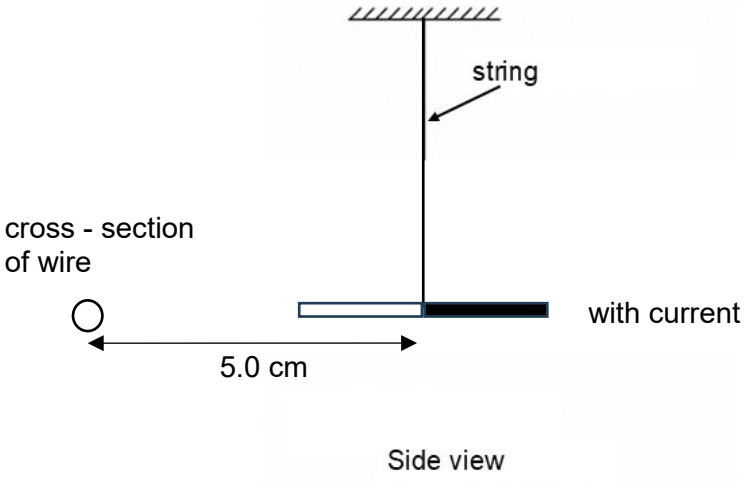


Fig. 5.4

(i) Show that the magnetic flux density produced by the wire, when the current is flowing through it, at the position of the short magnetic needle is 1.9×10^{-6} T.

The magnetic field due to a long straight wire is

$$B_{\text{wire}} = \frac{\mu_0 I}{2\pi r}$$

$$= \frac{(4\pi \times 10^{-7})(0.47)}{2\pi(0.050)}$$

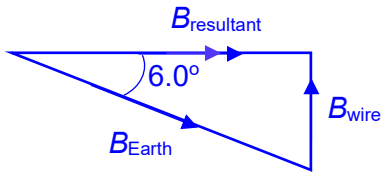
$$= 1.88 \times 10^{-6} \text{ T} \quad [1]$$

$$\approx 1.9 \times 10^{-6} \text{ T} \quad [1]$$

(ii) Determine the magnitude of the Earth's magnetic flux density in the laboratory.

The Earth's magnetic field makes an angle of 6.0º below the horizontal.
The wire's field is vertical at the position of the needle, so for the needle to become horizontal, the vertical component of the Earth's field must be cancelled.

$$B_{wire} = B_{Earth} \sin 6.0^\circ \qquad [1]$$



$$B_{Earth} = \frac{1.88 \times 10^{-6}}{\sin 6.0^\circ}$$
$$= 1.80 \times 10^{-5} \text{ T} \qquad [1]$$

magnetic flux density = ...T [2]

Total: [9]

9 **(a)** Distinguish between *electromotive force e.m.f.* and *potential difference p.d.* using energy considerations.

E.m.f. is the electrical energy converted from other forms of energy per unit charge delivered by the source. *P.d.* is the electrical energy converted to other forms of energy per unit charge when charges pass from one point to another. [1]

(b) Fig. 9.1 shows a variable resistor and a fixed resistor R_1 connected to a battery with e.m.f E and internal resistance r. The variable resistor takes a range of values between 0 Ω to R_{max}.

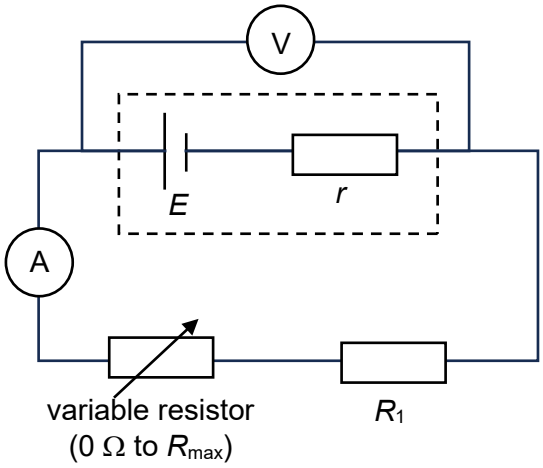


Fig. 9.1

The variation of the voltmeter reading V with current I through the ammeter as the variable resistor is varied over its **full range** of values is shown in Fig. 9.2.

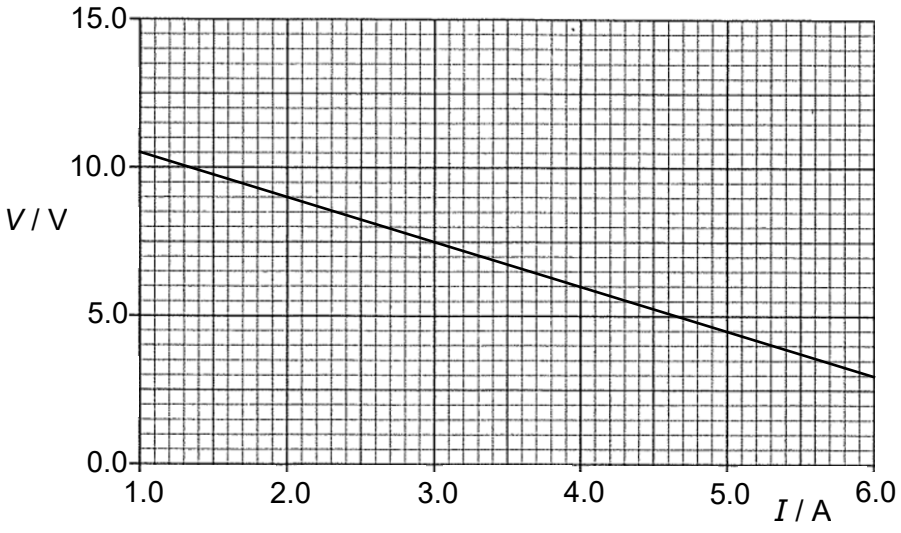


Fig. 9.2

(i) Using the graph, show that that internal resistance r is 1.5 Ω.

Since $E = V + Ir$ → $V = -rI + E$ (in the form of y = mx + c)

gradient of V–I graph gives **negative** value of internal resistance r

[1 identify which part of graph is related to r]

gradient = (10.5 – 3.0)/(1.0–6.0) [1 for correct coordinates on line, far apart]

gradient = –1.5 Ω

$\therefore r$ = 1.5 Ω [1]

[3]

(ii) Calculate the e.m.f. E of the battery.

When V = 10.5 V, I = 1.0 A

Using $E = V + Ir$

= 10.5 + 1.0(1.5) = 12.0 V

Note: E is **not** 10.5 V, as current at that point is not 0 A. Hence not y-intercept

e.m.f. E = 12.0 V [1]

(iii) Using Fig. 9.2, calculate the maximum resistance of the variable resistor R_{max}. Explain your working.

When variable resistor is set to 0 Ω, the resistance in the circuit is the lowest, hence current will be the highest. V = 3.0 V, I = 6.0 A

p.d. across R_1 is equal to V.

Using $V = IR_1$,

$R_1 = V / I$ = 3.0/6.0 = 0.50 Ω [1]

When variable resistor is set to max, the current will be the lowest.
V = 10.5 V, I = 1.0 A ,

Using $V = IR$, where R is the total resistance of R_{max} and R_1

$R_{max} + R_1 = V / I$

R_{max} + 0.50 = 10.5 / 1.0

R_{max} = 10.0 Ω [1]

(1 mark for explanation of choices of points)

resistance R_{max} = ... Ω [3]

(c) The variable resistor is now set to 6.0 Ω, and a thermistor is connected in parallel with the variable resistor and R_1. The ammeter is also now connected in series with the thermistor as shown in Fig. 9.3.

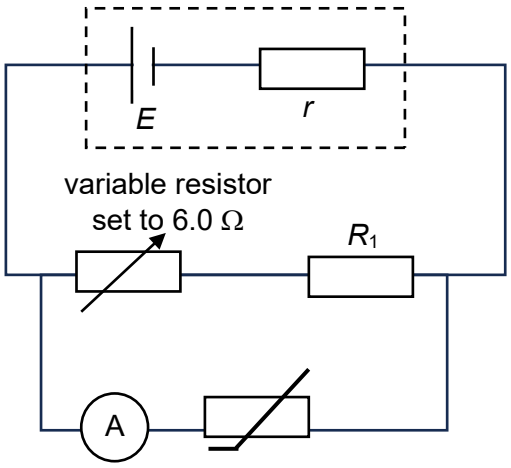


Fig. 9.3

(i) Explain how the potential difference across the thermistor changes when the temperature rises.

When the temperature increases, the resistance of the thermistor decreases. [1]

Hence the effective resistance of the external resistors and the thermistor, which are in parallel, decreases. By potential divider principle, the potential difference across both the external resistor and thermistor decreases. [1]

[2]

(ii) Hence, or otherwise, determine how the current across the thermistor changes when the temperature rises.

Since the effective resistance of the two external resistors and the thermistor decreases, the total resistance of the circuit decreases. Which means that the total current I_{total} exiting the cell increases.[1] Since the potential difference across the two external resistors decreases, by $V=IR$ the current I_R across the two external resistors decreases. [1]

Since $I_{total} = I_R + I_{Thermistor}$, the current through the thermistor increases. [1]

[3]

(d) Fig. 9.4 shows an alternating input voltage V_{IN} connected to a capacitor C and a resistor R.

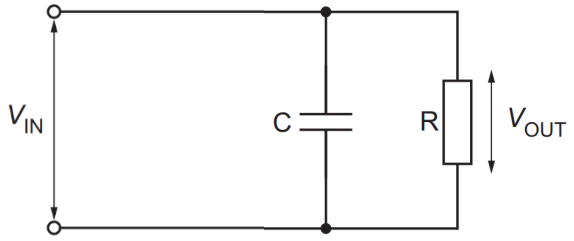


Fig. 9.4

The input voltage V_{IN} is a square wave. Fig 9.5 shows the variation of V_{IN} with time t.

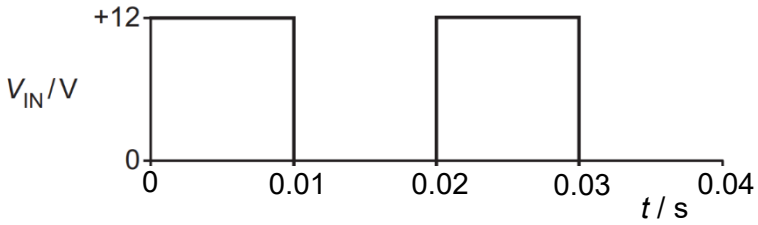


Fig. 9.5

Fig. 9.6 shows the variation of V_{OUT} with time t.

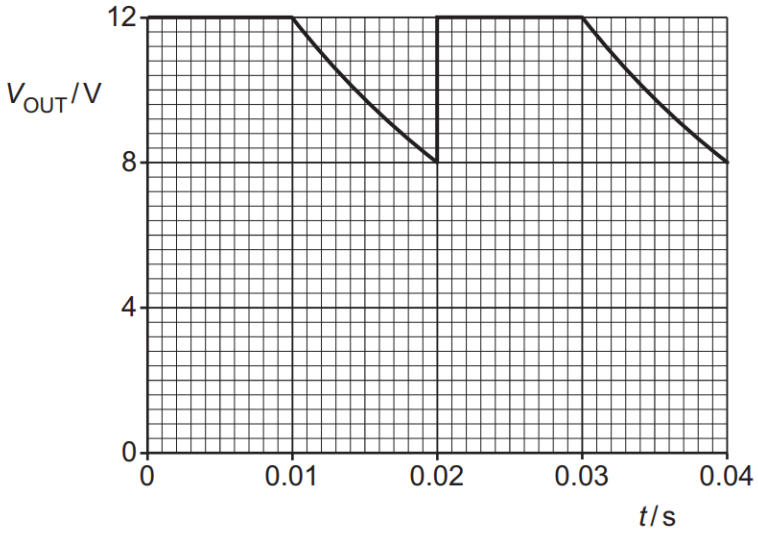


Fig. 9.6

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is 570 μF.

Energy stored in a capacitor $E = \frac{1}{2}CV^2$

$C = 2E/V^2 = \frac{2 \times 0.041}{12^2}$ [1]

$C = 569.44 \times 10^{-6}$ [1]

$= 570\ \mu F$

[2]

(ii) Determine the resistance of R.

It takes 0.010 s for p.d. to drop from 12.0 V to 8.0 V

Using $V = V_0 e^{-\frac{t}{\tau}}$, where $\tau = RC$

$8.0 = (12.0)\, e^{-\frac{0.010}{RC}}$ [1]

$\ln(8.0/12.0) = -\dfrac{0.010}{R \times 570 \times 10^{-6}}$

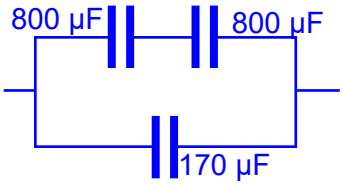
$R = 43.27\ \Omega = 43\ \Omega$ [1]

(note1: while not a marking pt, time is read to half a square precision, 0.010s)

(note2: recommended to use the proven value of C)

resistance = ... Ω [2]

(iii) Given two capacitors rated 800 μF and a third capacitor of 170 μF, suggest how capacitor *C* of 570 μF can be constructed from these three capacitors. Include a labelled diagram and show your workings clearly.



1m for diagram

Capacitance of two 800 μF in series = $\left(\dfrac{1}{800} + \dfrac{1}{800}\right)^{-1} = 400$ μF [1]

Capacitance of 400 μF and 170 μF in parallel = 400 + 170 = 570 μF [1]

[3]

[Total: 20]